

Design for Manufacturability

UCSD ECE 260C Spring 2026

Prof. Andrew Kahng

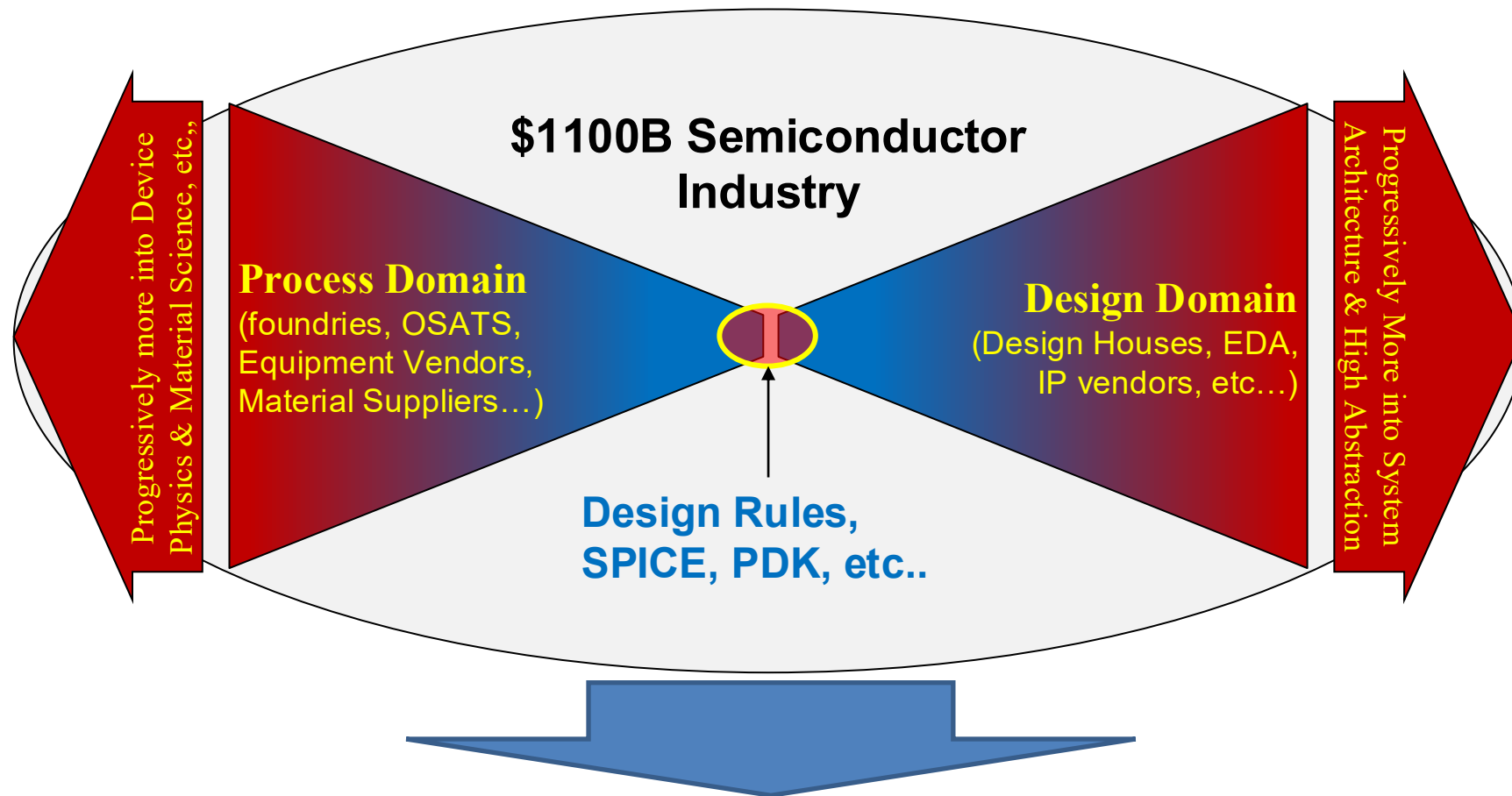
abk@ucsd.edu

Starting Point: “PPAC”

- **PPAC is why the industry continues “Moore’s Law scaling”**
 - Must see sufficient benefit from move to the next technology node
 - Moore’s Law = law of cost reduction (most straightforwardly, from scaling = integration)
- **Power, Performance, Area**
 - Dynamic/Static power, SAIF/VCD, ...
 - Incremental STA-driven sizing + buffering + synthesis operations
 - Better floorplans, better P&R tools, better “Opt” (remove overdesign)
- **What about Cost?**
 - Yield = fraction of chips produced that are “good” (sellable)
 - Cost of chip: (cost of fabricating, testing a wafer) / (yield * #chips/wafer)
 - Want higher yield (**Manufacturability**); also want higher #chips/wafer (**Area**)
 - “good” means passing both **functional** and **parametric** criteria
 - Functional: works correctly
 - Parametric: is sellable (performs at spec, not too leaky, won’t fail in a week, ...)

DfX $X = \{\text{process, interactions, reliability, manufacturing, temperature, stress, multiphysics, ...}\}$

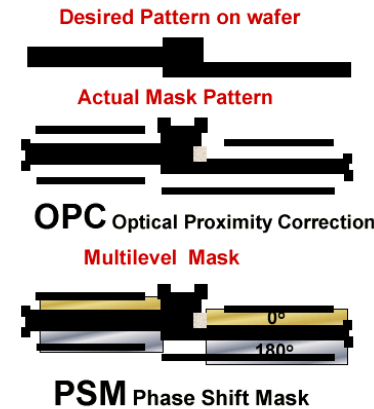
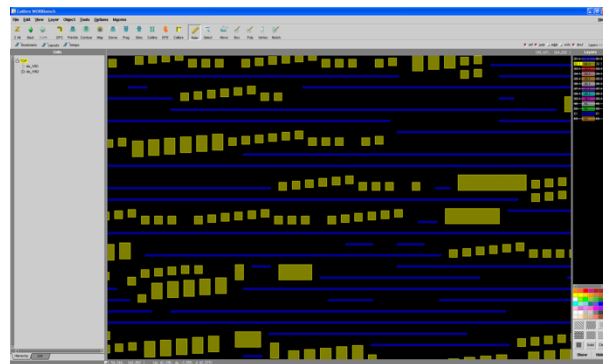
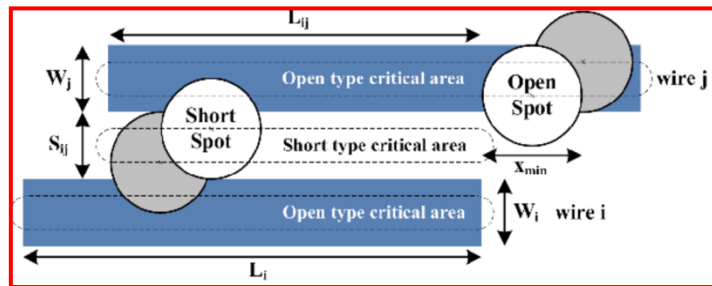
- \$1100B Semiconductor industry is divisible into Design-Centric and Process-Centric disciplines
 - *How do these disciplines communicate with each other?*
- Designers **experience** process technologies through **Design rules, SPICE models, PDK ...**



DfX = Succession of Methodologies that have shaped Design rules, SPICE models, PDK ...

DFM = 2-Way Street between “D” and “M” very hot topic in early 2000s

- **Geometric** fidelity: printed shape = drawn shape → geometric DFM
- **Electrical** fidelity: printed shape functions as designer assumed → electrical DFM
 - <https://vlsicad.ucsd.edu/Publications/Journals/j101.pdf>
- **Spread, fatten, fill**: part of “classic” DFM
- DFM impacts all axes of PPAC scaling
- DFM impacts “last levers” for scaling: quality, schedule, cost ($\approx$ value)



U.S. Patent Jun. 22, 2010 Sheet 9 of 10 US 7,743,349 B2

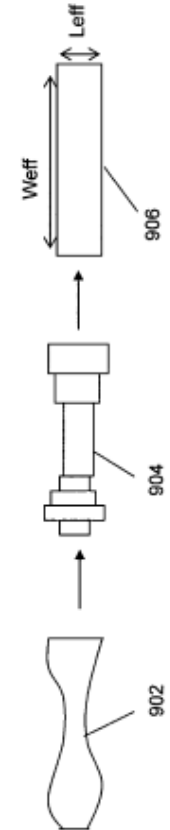
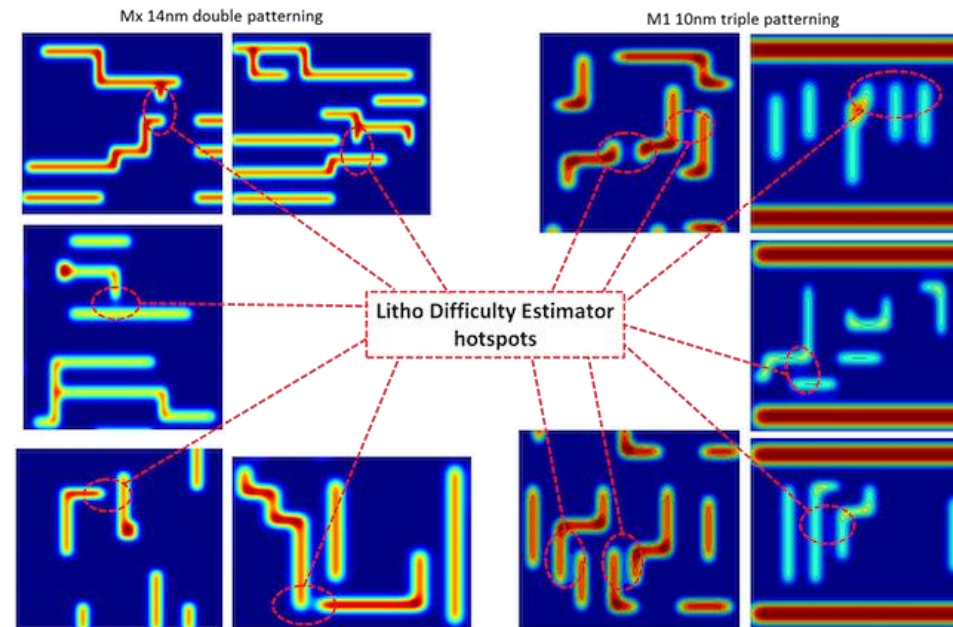


FIG. 9C

FIG. 9B

FIG. 9A

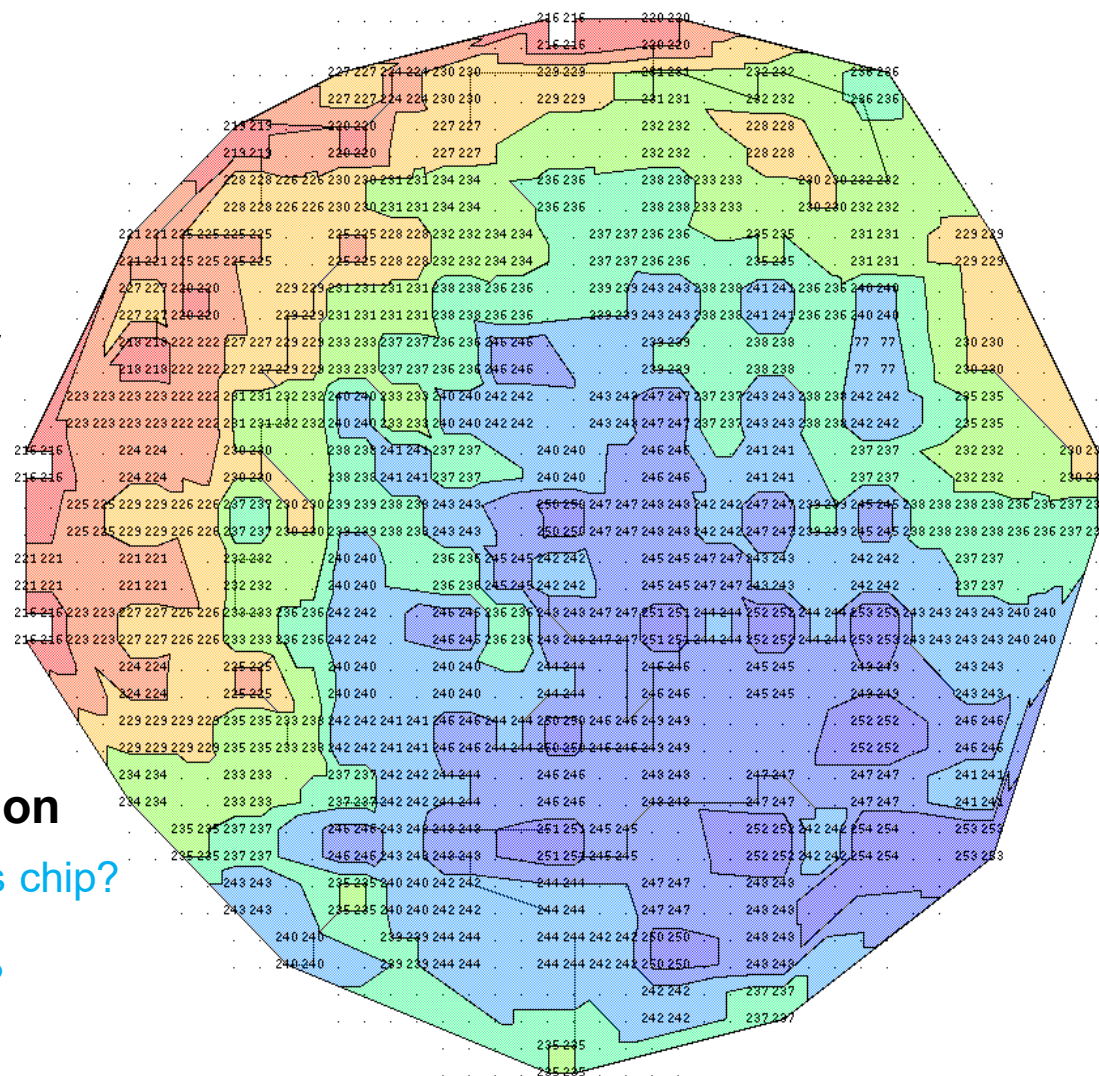


<https://www.techdesignforums.com/practice/technique/lithography-early-issue-detection-dtco/>

A Fundamental Problem: Unintentional Variations

- Chips don't work as designed
- Loss of predictability →
 - Guardbands
 - Overdesign
 - Worse time to market, worse cost, worse power
 - Loss of product value

Picture shows: Radial bias on wafer: bending, spin-on resist, ...
Sometimes, maximum use of wafer shots can be less profitable!
[Source: IBM Corp. ca. 65nm node]



Across-wafer frequency variation

What is the performance spec for this chip?

Is this part binned?

Does it have adaptivity mechanisms?

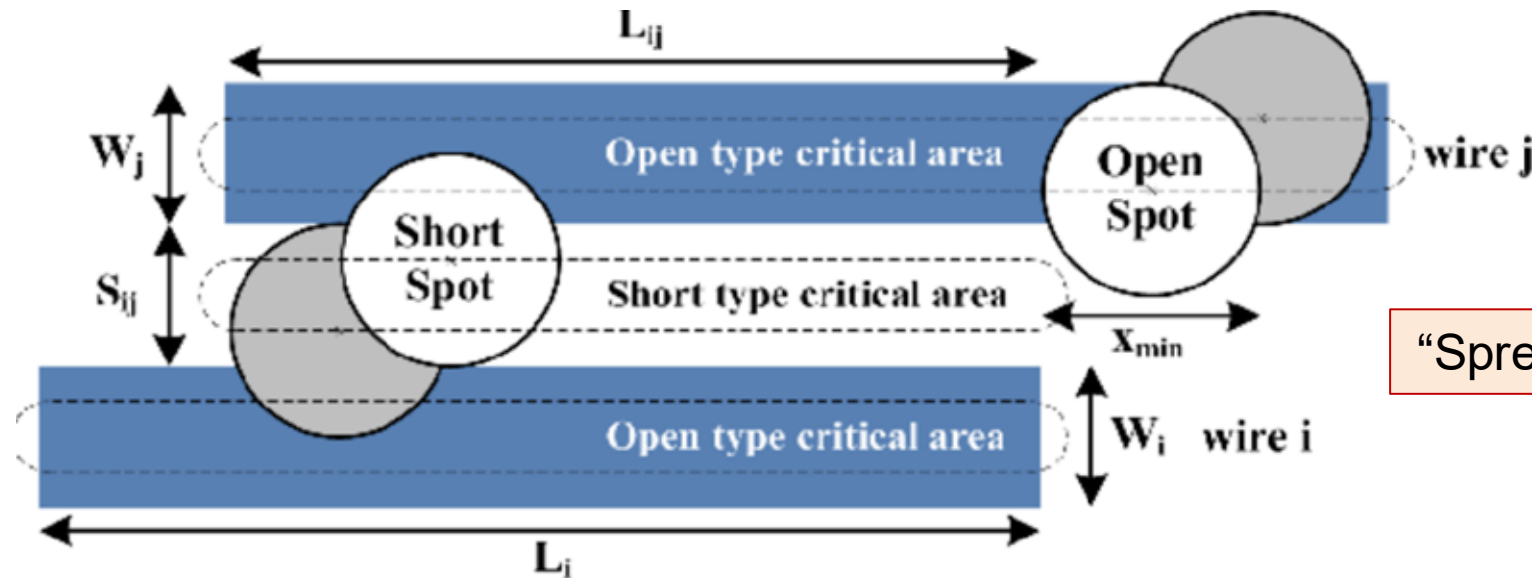
Etc. ...

Example Cost Arithmetic: Yield → Cost → Risk)

- Consider a chip selling 100M copies
 - Design house paid \$6K/300mm wafer in early 16nm technology
 - (Leading-edge wafer cost is higher now)
 - 10mm x 10mm die size → ~700 die/wafer
 - 90% vs. 95% yield
 - 630 vs. 665 good die per wafer
 - 158730 vs. 150370 wafers needed to meet the demand
 - Difference: \$50M + extra time to finish producing 100M chips
- What matters is *good die/wafer*
 - Not too slow, not too power-hungry....

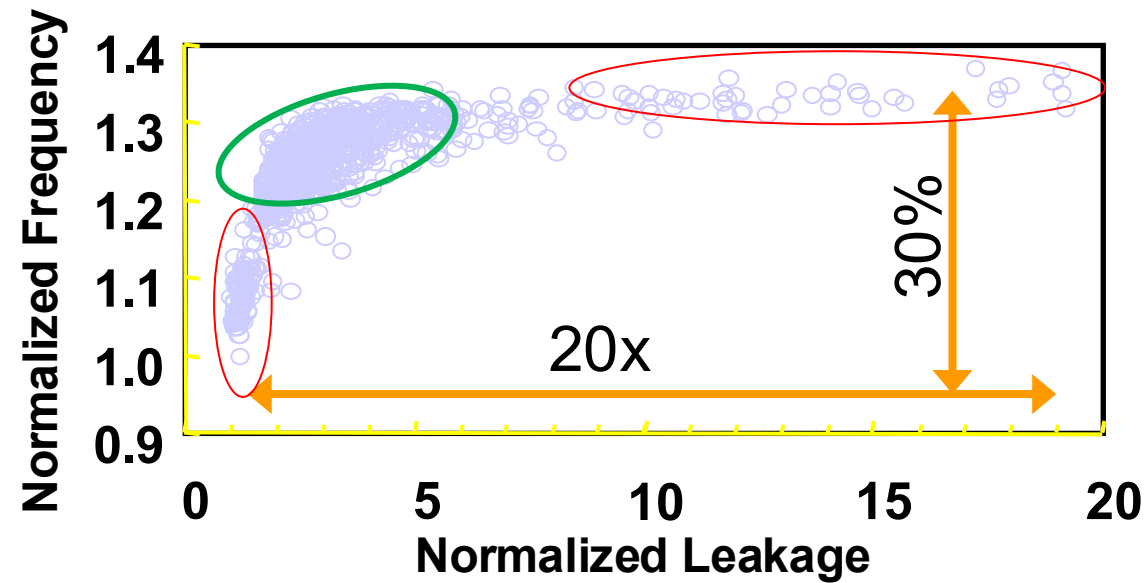
Functional Yield

- No physical faults: “opens”, “shorts” etc.
 - DFT, ATPG, ATE, ... spans EDA and Test supplier industries
- “Critical area” = vulnerability to particular defect sizes [see the dashed outlines](#)
- How much of my layout will suffer an “open” if an open spot of a particular size were to randomly appear?
 - Wider wires $\rightarrow$ layout is less vulnerable to “open” faults
- How much of my layout area will suffer a “short” if extra material of a particular size were to randomly appear?
 - Wider spaces $\rightarrow$ layout is less vulnerable to “short” faults



“Spread, Fatten and Fill”

Parametric Yield Example: Leakage Power Variability



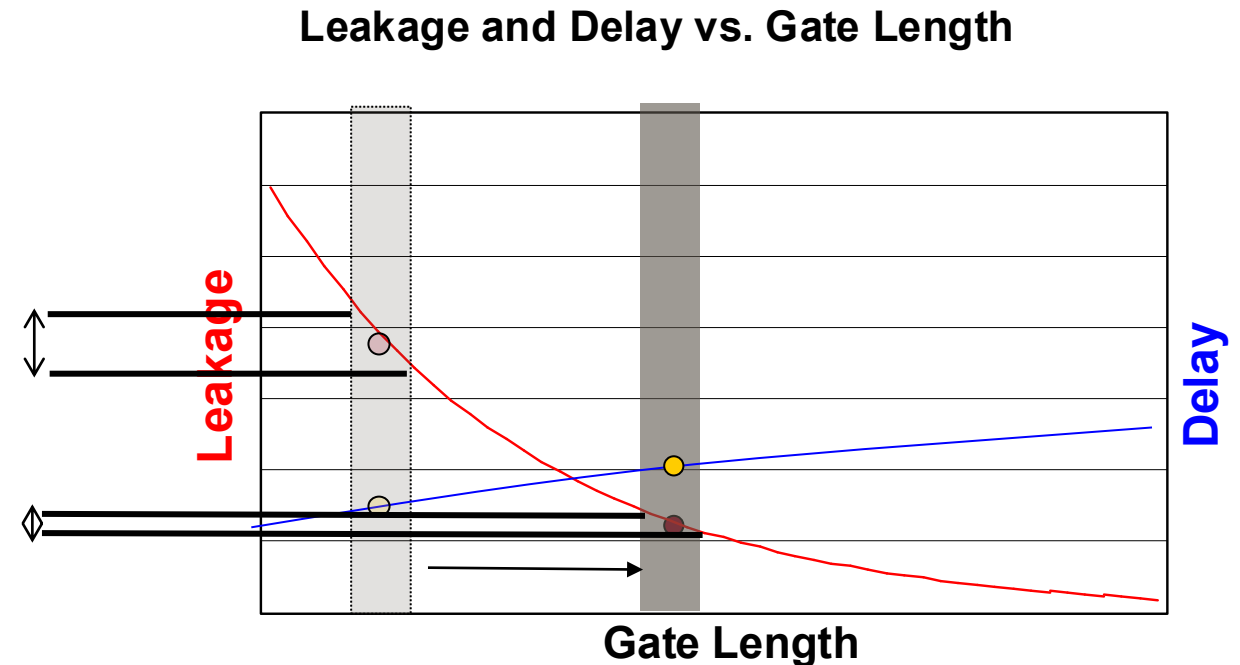
Source: Intel Corp.
ca. 130nm node

- Leakage power variability

- Subthreshold leakage exponential in almost everything (L , V_t , T_{ox} , Temperature, Voltage)
⇒ 5-20X variation is common
 - Gate length (= “ L_{gate} ”, or “CD” – “critical dimension”) manufacturing variation is biggest source
- **Problems:** power-limited yield loss, thermal runaway, inability to perform ‘burn-in’ reliability testing
- How does design help deal with this manufacturing-induced variation?

Concept: DFM Includes “Design-Aware Manufacturing”

- Transistor leakage and speed vary with channel length
 - Subthreshold leakage varies exponentially with gate length
 - Delay varies linearly with gate length
- “Gate-length biasing” (Multi-Lgate, CD biasing, Wimpy transistors, ...): Positive timing slack can be “converted” into power reductions (smaller transistors, area, power, ...)
- Larger gate lengths also have less leakage variation (!)
- Knob: transistor gate length ...
- But: how should this design-awareness be passed to manufacturing?



Gupta et al.

<https://vlsicad.ucsd.edu/Publications/Conferences/177/c177.pdf>

<https://vlsicad.ucsd.edu/Publications/Journals/j79.pdf>

UCSD Transistor Gate Length Biasing, early 2000s

- **Bias Impact**

- Exponential leakage, variability reduction
- Linear performance reduction

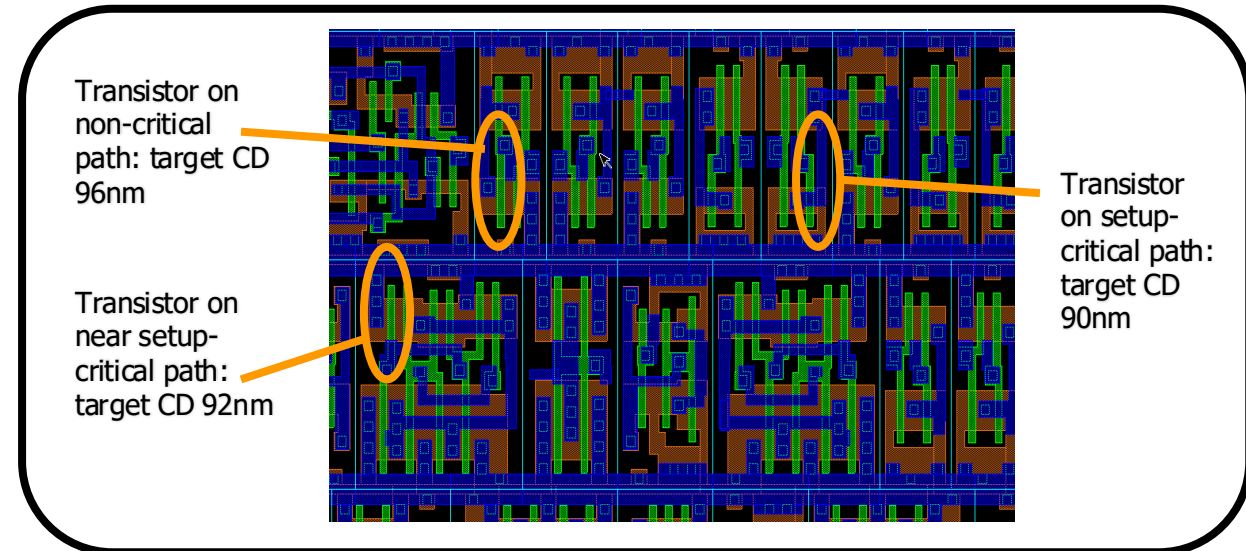
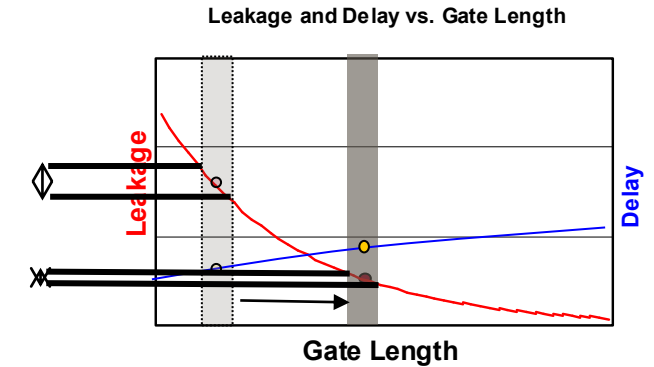
- **Idea: Use fine-grain biases (2nm, 3nm)**

- Small leakage reduction beyond 10% biasing
- **Preserves pin-compatibility** → Technique can be applied post-routing

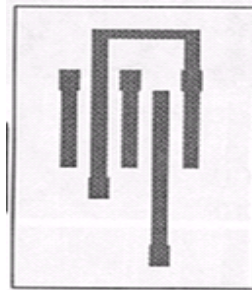
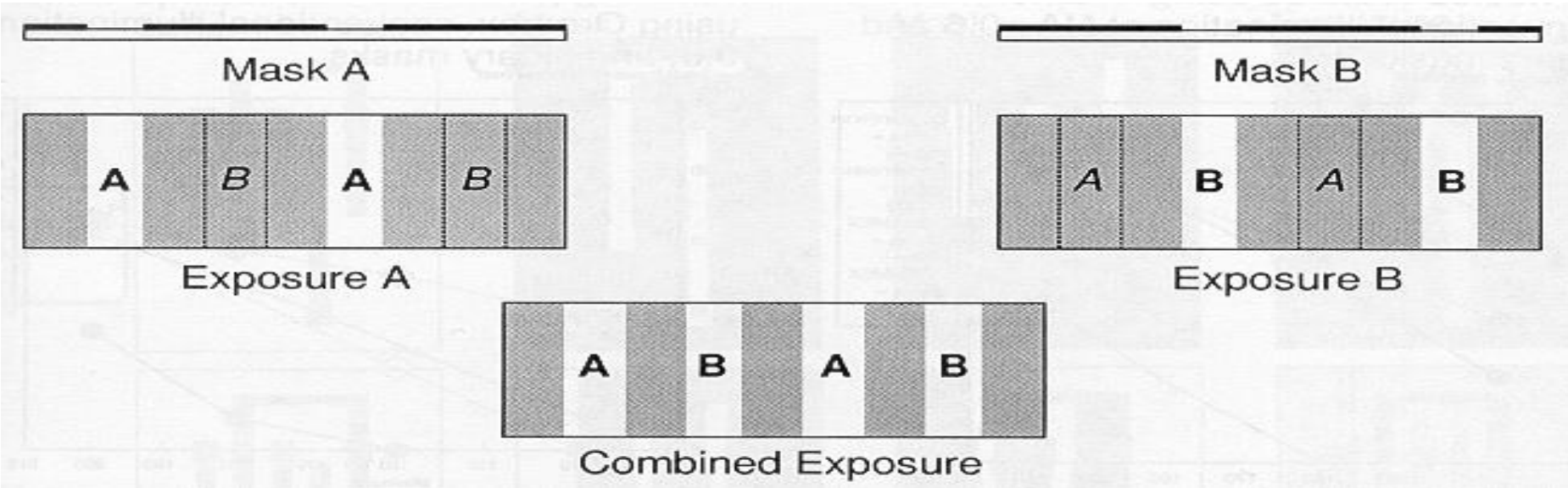
- **Design** benefit: Reduce leakage power by 20%, reduce leakage variability by 30%

- **Manufacturing** benefit: Same process offers more value to customer: power, speed, variability, parametric yield

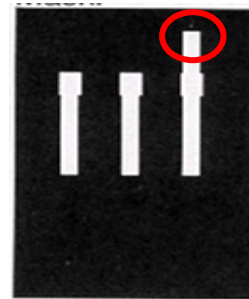
- (For some years: TSMC's "PowerTrim Service")



Multi-Patterning: e.g., Double-Patterning Lithography (DPL)

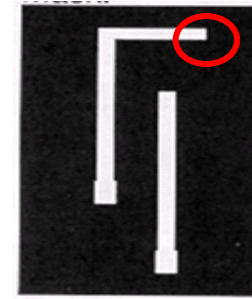


Desired pattern

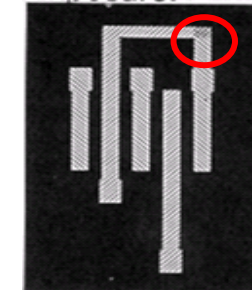


First Mask

+



Second Mask

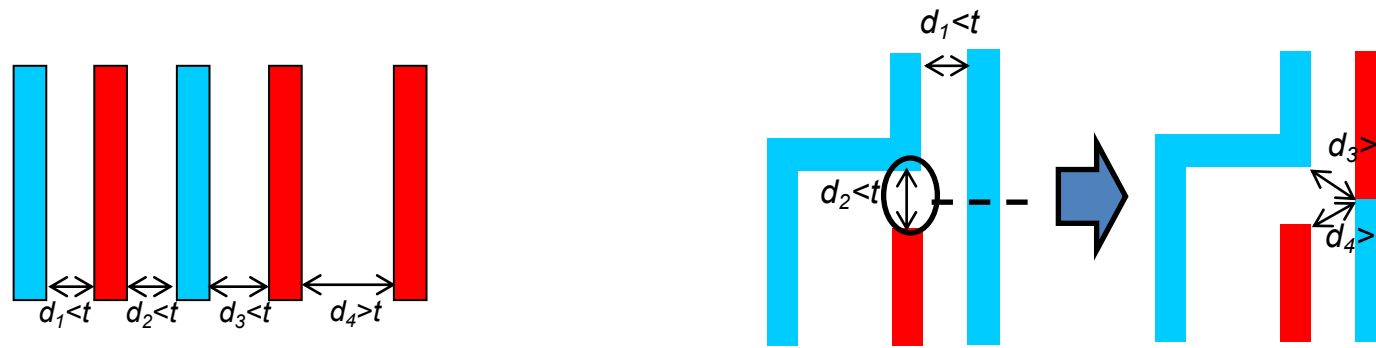


Combined exposure

“LELE” double-patterning: litho-etch-litho-etch

DPL Layout Decomposition

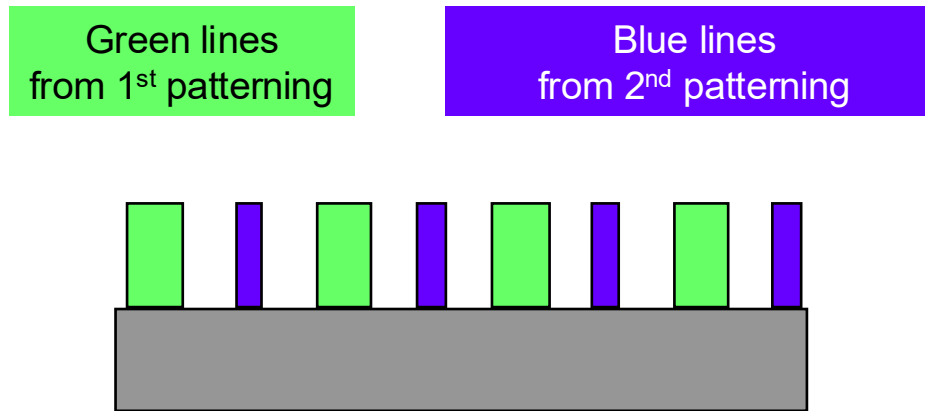
“stitch” of shapes on two separate masks, to make one feature



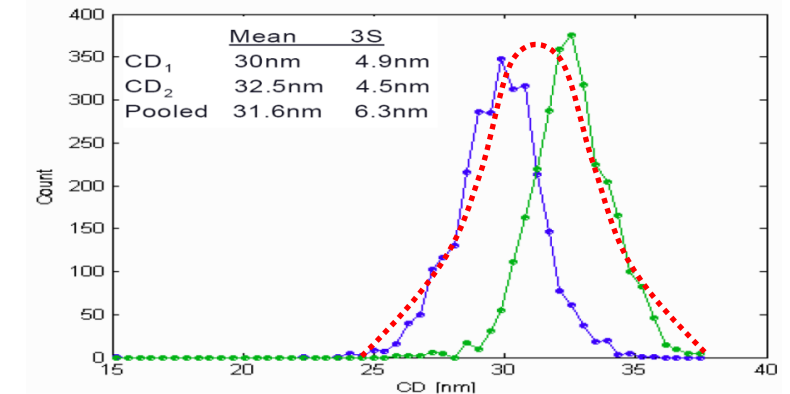
- Two features **must** be assigned opposite colors if their spacing is less than the minimum coloring spacing t
- IF two features within the minimum coloring spacing t cannot be assigned opposite colors
 - **THEN** at least one feature must be **split** into two or more parts
 - (Alternatively: We could move a feature or widen a feature ...) ← Not so great for “convergence”!
- Pattern **split** increases manufacturing cost, complexity
 - Line ends → corner rounding
 - Overlay error and interference mismatch → line edge errors → tight overlay control
 - Optimization: minimize cost of layout decomposition
 - “**Graph Bipartization**” Problem: Given a conflict graph that has odd-length cycles, “bipartize” it = make it 2-colorable by deleting edges, splitting nodes, or both ...

Concept: DFM Includes “Manufacturing-Aware Design”

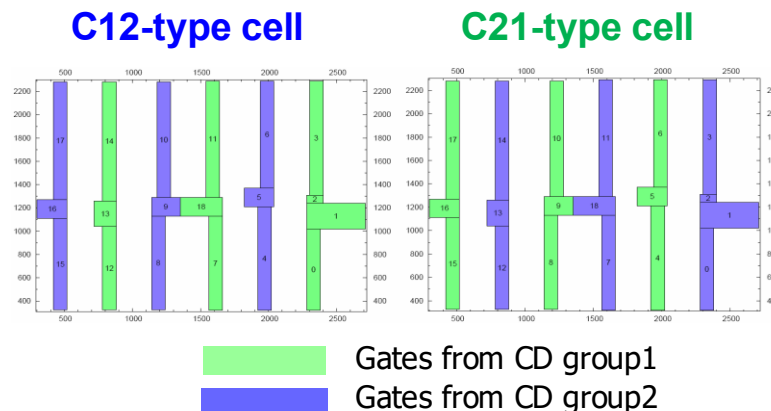
- Key Insight: Double-patterning induces “Bimodality”
- Two patterning steps → Two different CDs



“LELE” double-patterning: litho-etch-litho-etch



- Two different colorings → Two different timings !!! (for two instances of the same cell !!!)



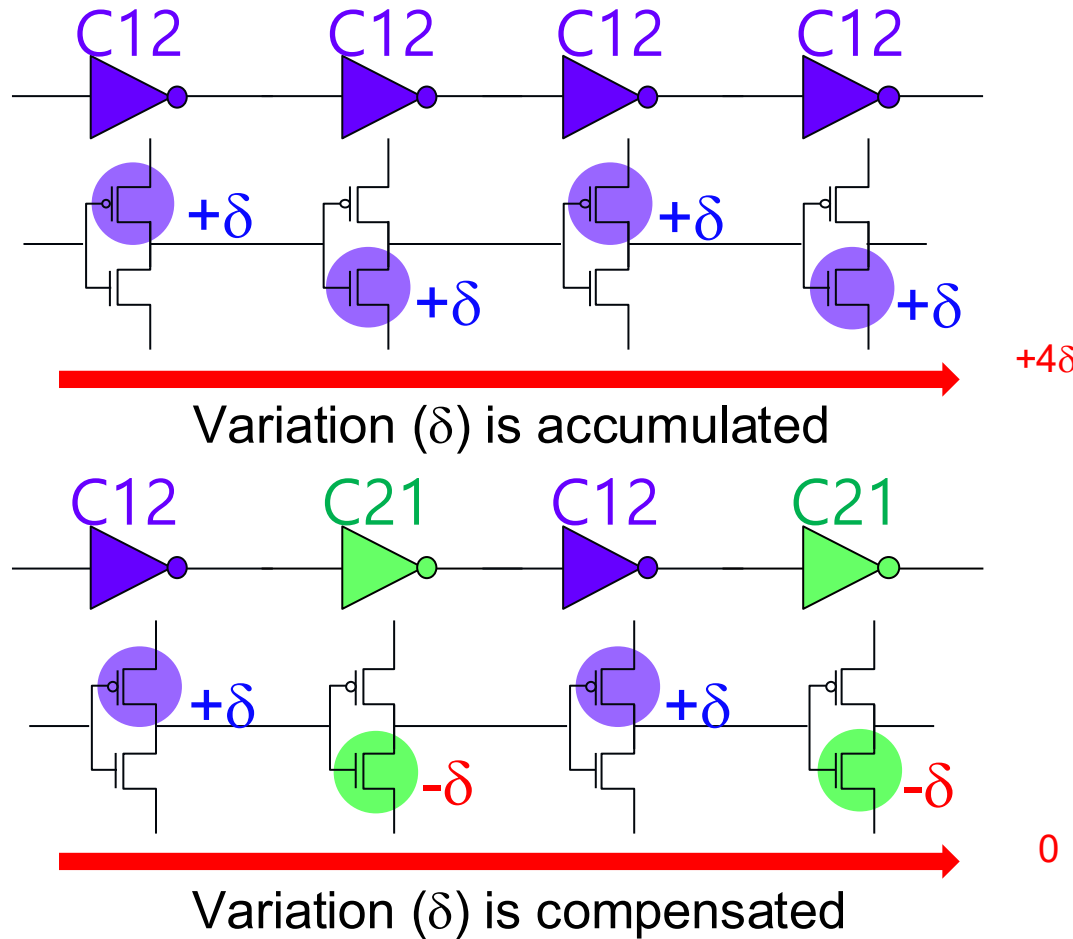
C12: Odd polys in BLUE,
Even polys in GREEN

C21: Odd polys in GREEN,
Even polys in BLUE

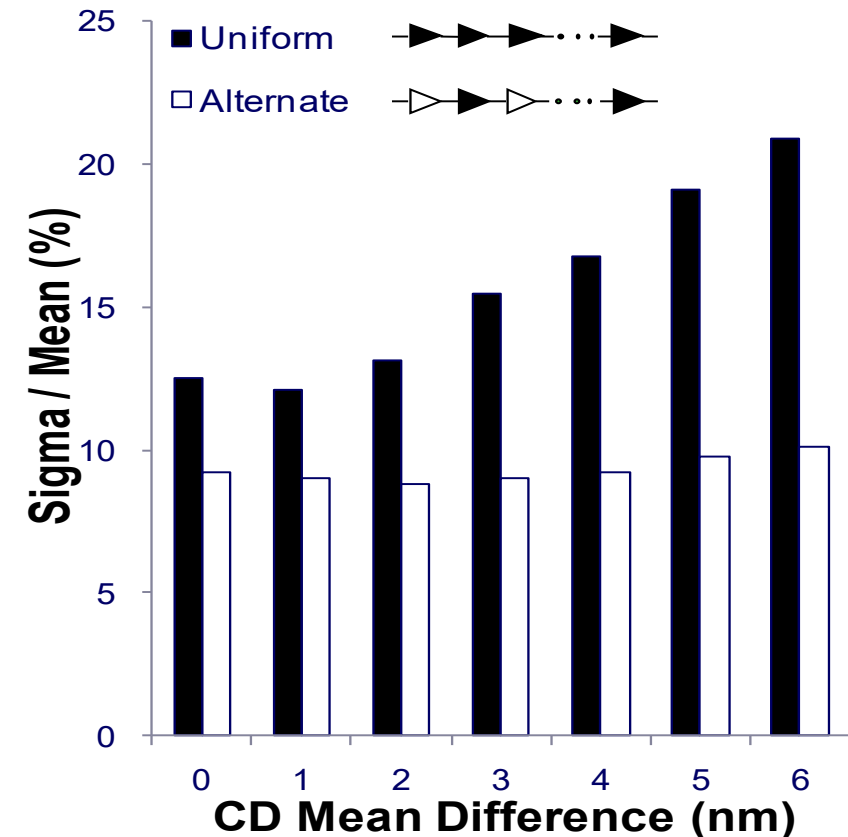
<https://visicad.ucsd.edu/Publications/Journals/j100.pdf>

Impact of Bimodality on Data Path Delay

- Bimodality can help reduce path delay variation
 - Reduction of covariance when alternately (i.e., mixed-) colored



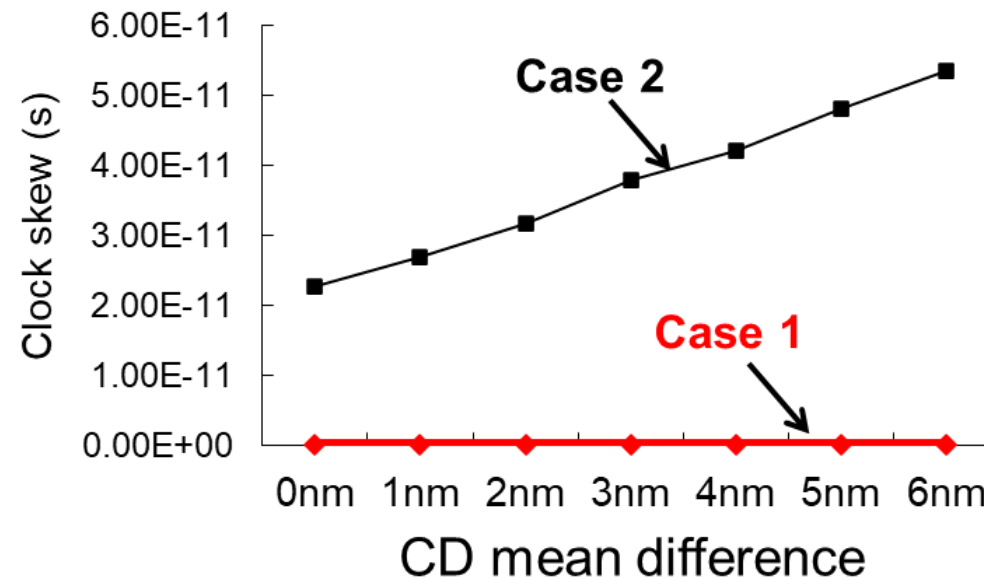
SPICE Simulation Results



Impact of Bimodality on Clock Skew

- Different combinations of colors in a clock tree → **can cause SKEW**

	Source to Sink A	Source to Sink B
Case 1	C12+C12+C12+...+C12	C12+C12+C12+...+C12
Case 2	C12+C12+C12+...+C12	C21+C21+C21+...+C21



- Same coloring on all clock buffers is better!**

Bimodal CD Distribution and Manufacturing-Aware Design

<https://visicad.ucsd.edu/Publications/Journals/j100.pdf>

1. Design requires **bimodal-aware timing models**

- Unimodal representation is too pessimistic

2. Data paths benefit from mixed coloring

- Exploit existence of two uncorrelated CD populations
- Minimize correlated variations in a given path

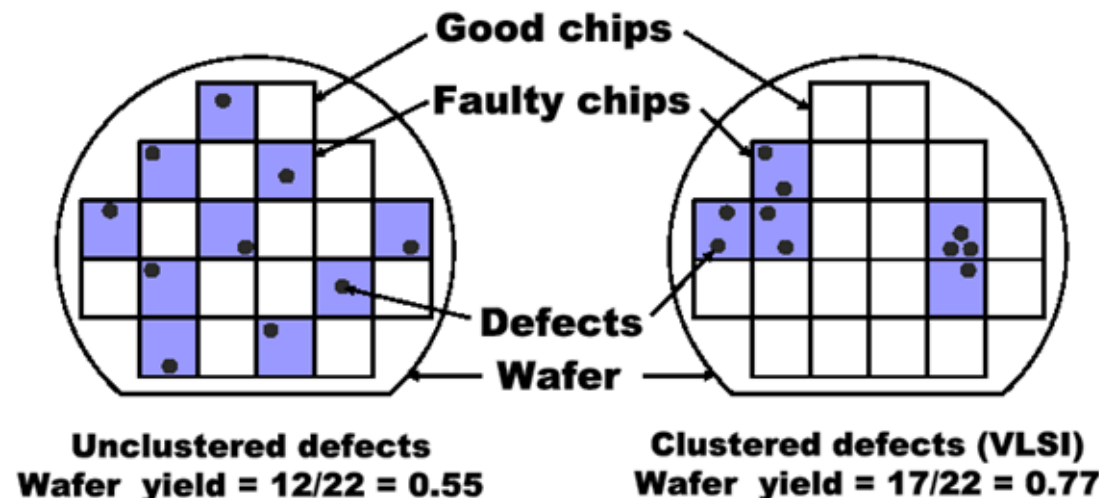
3. Clock paths benefit from uniform coloring

- Correlated variation between launch and capture paths minimizes bimodality-induced clock skew

⇒ **Manufacturing-Aware Design can exploit both correlated, uncorrelated variations!**

Enormous Engineering Efforts ...

- Modeling of defects
 - E.g., clustering
- Cleanroom technologies
 - E.g., factory layout, airflow, robots, ...
 - “Class 1 cleanroom” etc. [link](#)
- “Design for Manufacturability”
 - E.g., to first order, make chip smaller (~cubic yield loss w/area)
 - Better layouts: “Spread, fatten, fill”, avoid risky patterns, follow “recommended rules” aka “DFM rules”, swap in “DFM cells”, ...
 - Redundancy / error tolerance
 - Doubled vias and contacts
 - Error detection/correction in memories
 - Better OPC / mask flows



Note: hard to cleanly separate timing, power, reliability, yield ...

OpenROAD Task 12: Layout Finishing

- “The goal of Task 12 is to ensure compliance with requirements for manufacturing handoff, and to accomplish post-layout optimizations for manufacturability.”
- Objective:** Develop layout finishing functions for incorporation in layout generation tools.
- Completion Criteria:** *Demonstration of rules-compliant layout finishing functions including TCD, well taps, antenna checks, GDS merge, dummy fill.*



Microsystems Technology Office
Broad Agency Announcement
Electronics Resurgence Initiative: Page 3 Investments
Design Thrust
HR001117S0054
September 15, 2017
Amendment No. 01
As Amended on September 19, 2017

Table 2. Table identifying and describing subtasks proposers may propose to as part of IDEA TA-1. Performers may propose solutions for one or multiple of the identified subtasks.

	TA-1 Subtask	Description
Common Infrastructure	Databases / Processing	Create critical infrastructure for TA-1 performers, including but not limited to: distributed and in-memory design databases, infrastructure for distributed EDA computing on data-center and supercomputer scale computers.
	Cloud infrastructure	Develop compute infrastructure, IP firewalling, and merchant eco-system needed to allow the “no human in the loop” IDEA layout generator to be effectively accessed as a cloud based software service by the broader community.
	Timing Analysis	Full circuit and incremental scalable timing analysis software for synchronous and asynchronous circuits. Interacts with synthesis, floorplanning, placement, clock-tree synthesis, routing, and layout finishing sub-tasks through an open API.
	Parasitic Extraction	Extraction technology, foundry-specific parasitic models for capacitance, resistance, inductances. Interoperable with all TA-1 software modules.
	Readers + Writers	Optimized parsers, readers, writers for standard file formats such as Verilog, GDSII, LEF, DEF, SPICE, LIBERTY, SDC, and PDKs.
	Power and Signal Integrity Analysis	Circuit power and signal integrity software with support for instantaneous dynamic power analysis, static power consumption, electro-migration, reliability, and thermal analysis. Interacts with synthesis, floorplanning, placement, clock-tree synthesis, routing, and layout finishing sub-tasks through an open API.
Layout Generators	Logic Synthesis	Transform Verilog RTL to a gate level netlist (<u>digital circuits only</u>).
	Floor-planning	Generate correct by construction floor-plans using structured netlists, PDK, minimal user constraints, and trained models as inputs.
	Placement	Place all cells/parts using floorplan, structured netlist, PDK, minimal user constraints, and trained models as inputs.
	Clock Tree Synthesis	Synthesis and layout of a clock distribution network (<u>digital circuits only</u>).
	Detailed Routing	Route all nets in the design using placed circuit, PDK, minimal user constraints, and trained models as inputs.
	Layout Finishing	Design For Manufacturing optimizations using routed circuit, PDK, minimal constraints, and trained models as inputs.
Design	SoC Design Advisors	Expert design teams that will act as advisors to the IDEA layout platform developers and use the IDEA software to demonstrate “no human in the loop” layout of an appropriate 100mm ² SoC at 14nm CMOS as well as accompanying package and board.



OpenROAD's Task 12 (“DFM”)

Completion Criteria: *Demonstration of rules-compliant layout finishing functions including TCD, well taps, antenna checks, GDS merge, dummy fill.*

Tapcell Insertion (well taps) <https://teamvlsi.com/2020/08/well-tap-cell-in-asic-design.html>

- Tap cells prevent latch-up in CMOS
- Low-impedance path between power supply rails
- Taps connect (clamp) nwell to VDD, p-sub to VSS so parasitic transistors cannot turn on
- Modern cell libraries have “tapless” cells
 - No internal well connections → tapcells must exist within, e.g., 25um of any transistor
- <https://openroad.readthedocs.io/en/latest/main/src/tap/README.html>

Tapcell (Well Tap) Insertion

Verilog
+ libraries,
constraints

Logic Synthesis

Floorplanning

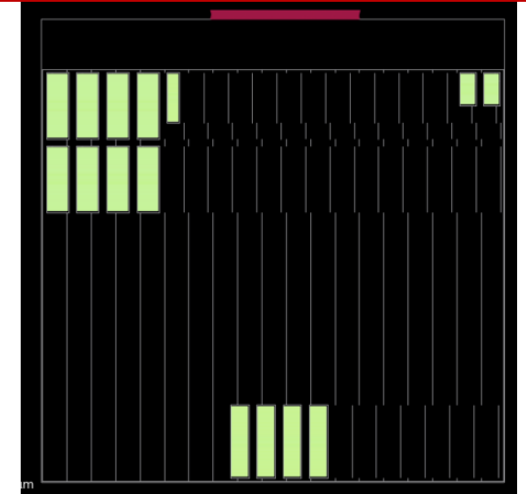
Placement

Clock & Optimization

Global and Detailed
Routing

GDSII
final
layout

Layout Finishing



UCSD ECE 260C SP26 Kahng

16

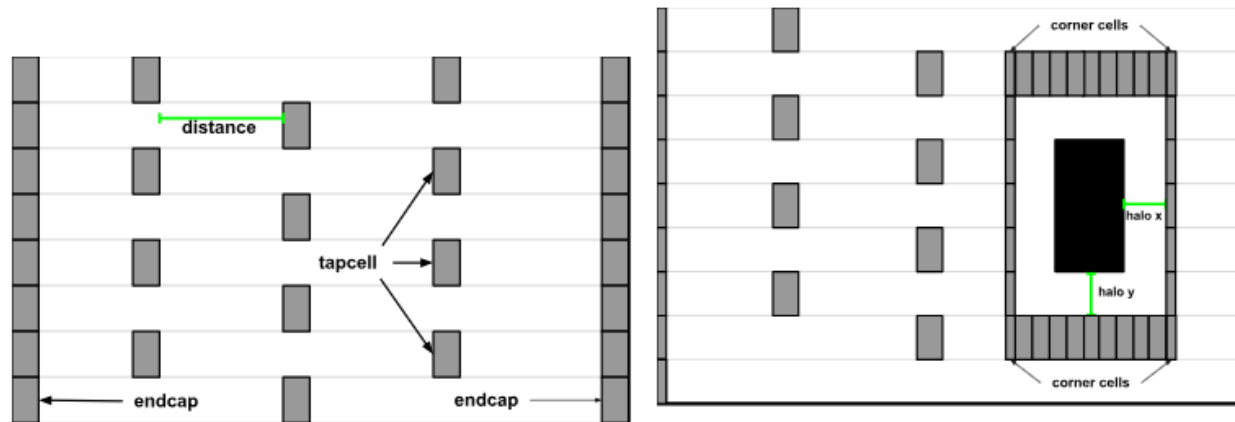


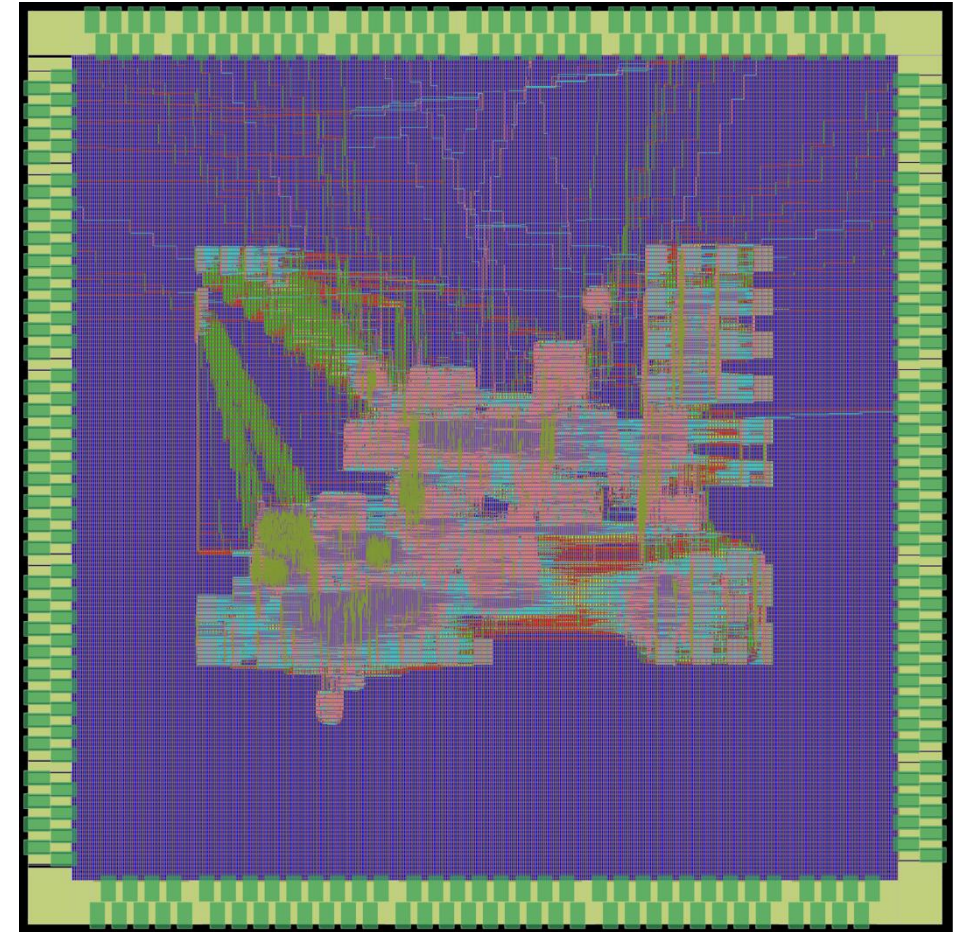
Figure 1: Tapcell insertion representation

Figure 2: Tapcell insertion around macro representation

```
tapcell
[ -cnrcap_nwin_master cnrcap_nwin_master ]
[ -cnrcap_nwout_master cnrcap_nwout_master ]
[ -disallow_one_site_gaps ]
[ -distance dist ]
[ -endcap_cpp endcap_cpp ]
[ -endcap_master endcap_master ]
[ -endcap_prefix endcap_prefix ]
[ -halo_width_x halo_x ]
[ -halo_width_y halo_y ]
[ -incnrcap_nwin_master incnrcap_nwin_master ]
[ -incnrcap_nwout_master incnrcap_nwout_master ]
[ -no_cell_at_top_bottom ]
[ -row_min_width min_width ]
[ -tap_nwin2_master tap_nwin2_master ]
[ -tap_nwin3_master tap_nwin3_master ]
[ -tap_nwintie_master tap_nwintie_master ]
[ -tap_nwout2_master tap_nwout2_master ]
[ -tap_nwout3_master tap_nwout3_master ]
[ -tap_nwouttie_master tap_nwouttie_master ]
[ -tap_prefix tap_prefix ]
[ -tapcell_master tapcell_master ]
[ -tbtie_cpp tbtie_cpp ]
```


TCD = Test-key Critical Dimension Cells <https://vlsibasic.blogspot.com/2017/01/tcd-cell.html>

- TCD = Test-key Critical Dimension Cell allows monitoring of CDU = CD uniformity
- E.g., in first-generation FinFET node (16nm-12nm) a TCD site would be responsible for $\sim 3 \text{ mm}^2$ of silicon area in which it was centered
- **This impacts the floorplan !**
 - E.g., need to have TCD sites on roughly a $\sim 1.7 \text{ mm}$ grid
- “BlackParrot” RISC-V core example: place four TCD cells into corners, to avoid disturbing floorplan



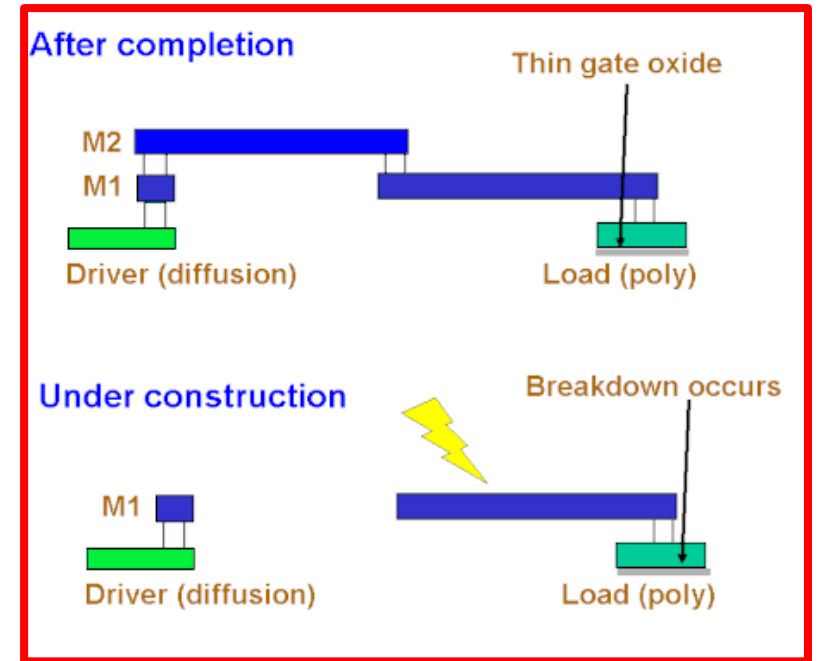
This is a 3mm x 3mm die. If each TCD cell can cover 3 mm^2 , how many would you place in the layout, at what locations?

Antenna Rules

https://en.wikipedia.org/wiki/Antenna_effect

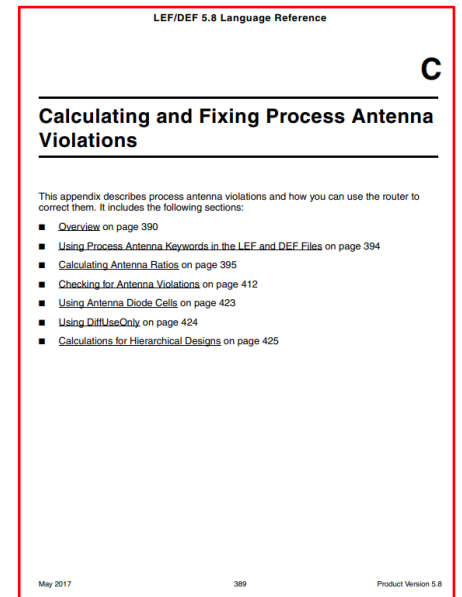
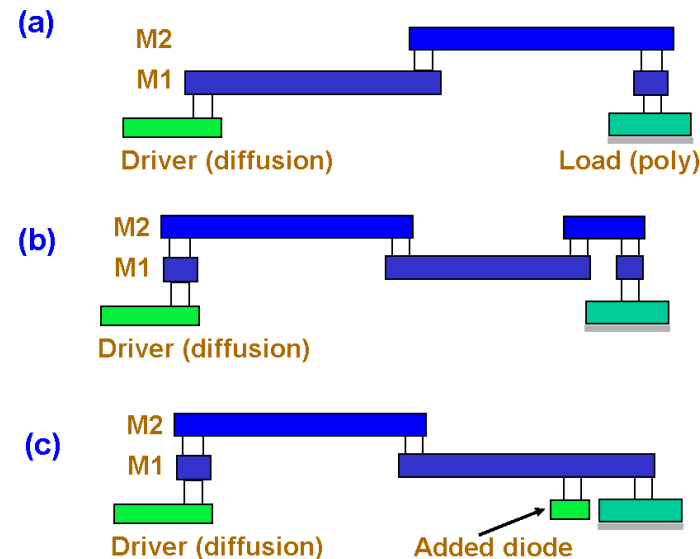
aka plasma-induced gate oxide damage

- In a signal net:
 - Driver** contains a source or drain diffusion
 - Sink** has a gate electrode over a thin gate dielectric (i.e., gate oxide)
 - After fabrication**, the net cannot be at higher voltage than VDD since each net has connected source/drain implant → diode breaks down at lower voltage than gate oxide
- During fabrication** (M1 etch), M2 does not exist yet, so stray charge can damage gate oxide of load



- Antenna rules limit ratio of “antenna area” to gate area**
(per-layer or cumulative area rules; perimeter can also be used in calculation)

- Fix via methodology or via EDA tool
 - Change order of routing layers, insert “bridges” = (a), (b) ← need smart DRT
 - Add protection diode = variant of sink cell, or special diode cell (c) ← disrupts layout, creates ECOs **OpenROAD uses this approach**
- When do you determine that a fix is needed?
 - OpenROAD makes this determination after global routing, and inserts protection diodes

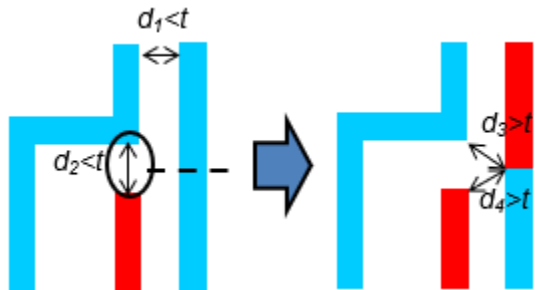


Checkpoint

- **TCD** handled as placed macro
- **Tapcell** command during floorplan initialization
- **Antenna check** and add protection diodes post-GR

Many workarounds in OR = opportunities to contribute ☺

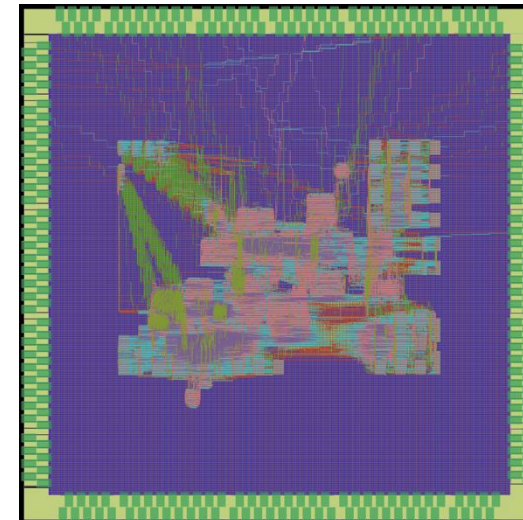
- **Double-patterning lithography?** → one-way routing today



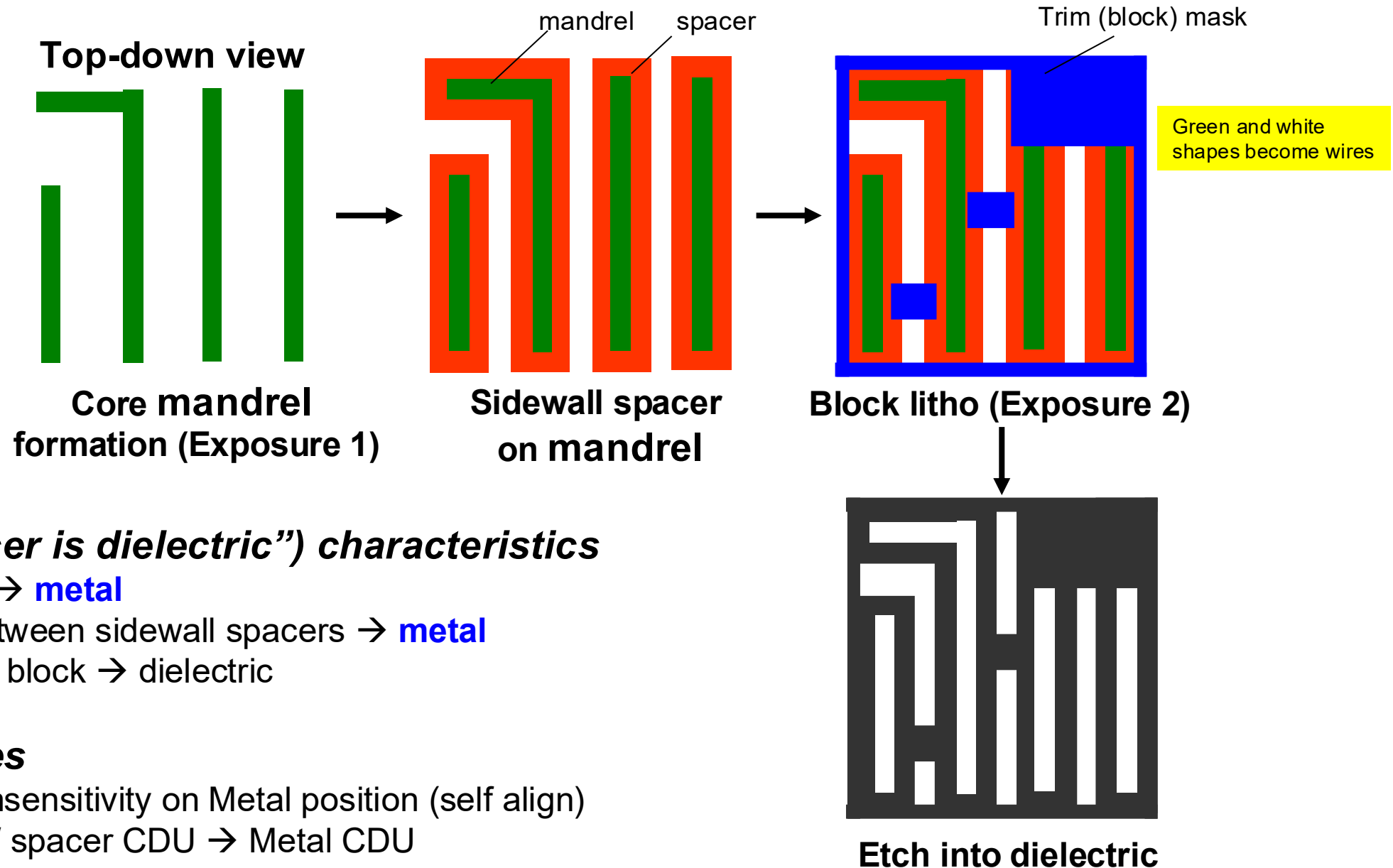
- **Lonely vias?** “Vx max space to Union (Vx, Vxbar, Vxlarge, ...) ≤ 4 → extend power mesh, shift floorplan, ... (and hope!)”

OpenROAD's Task 12 (“DFM”)

Completion Criteria: *Demonstration of rules-compliant layout finishing functions including TCD, well taps, antenna checks, GDS merge, dummy fill.*

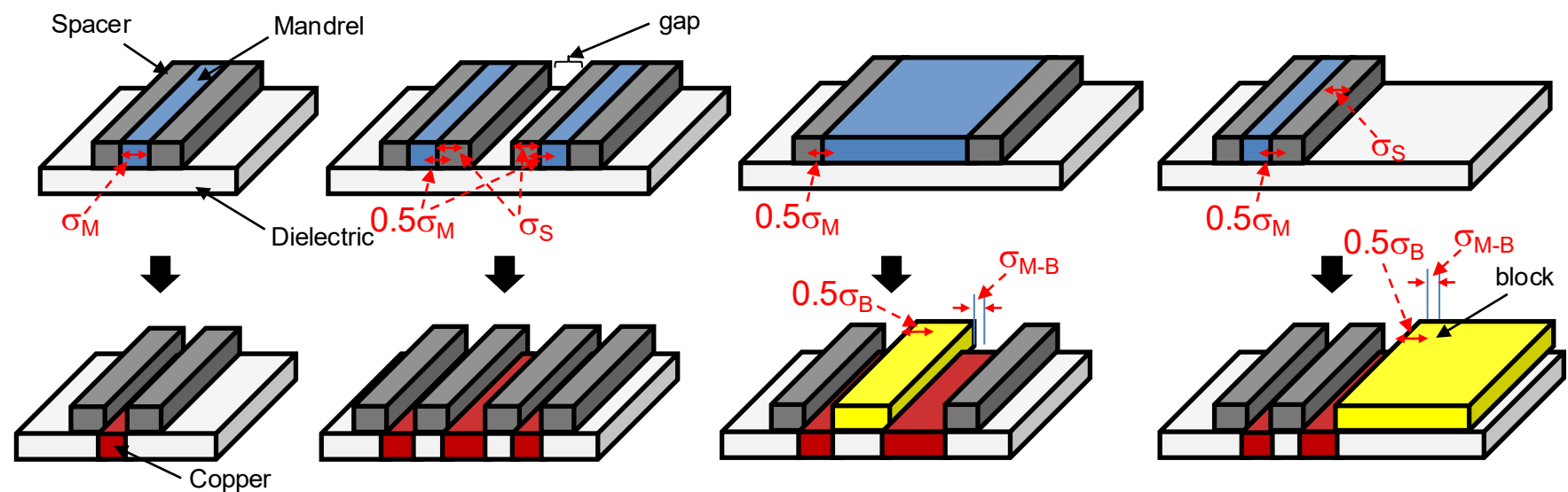


Checkpoint (cont.) – Spacer-Assisted Double-Patterning



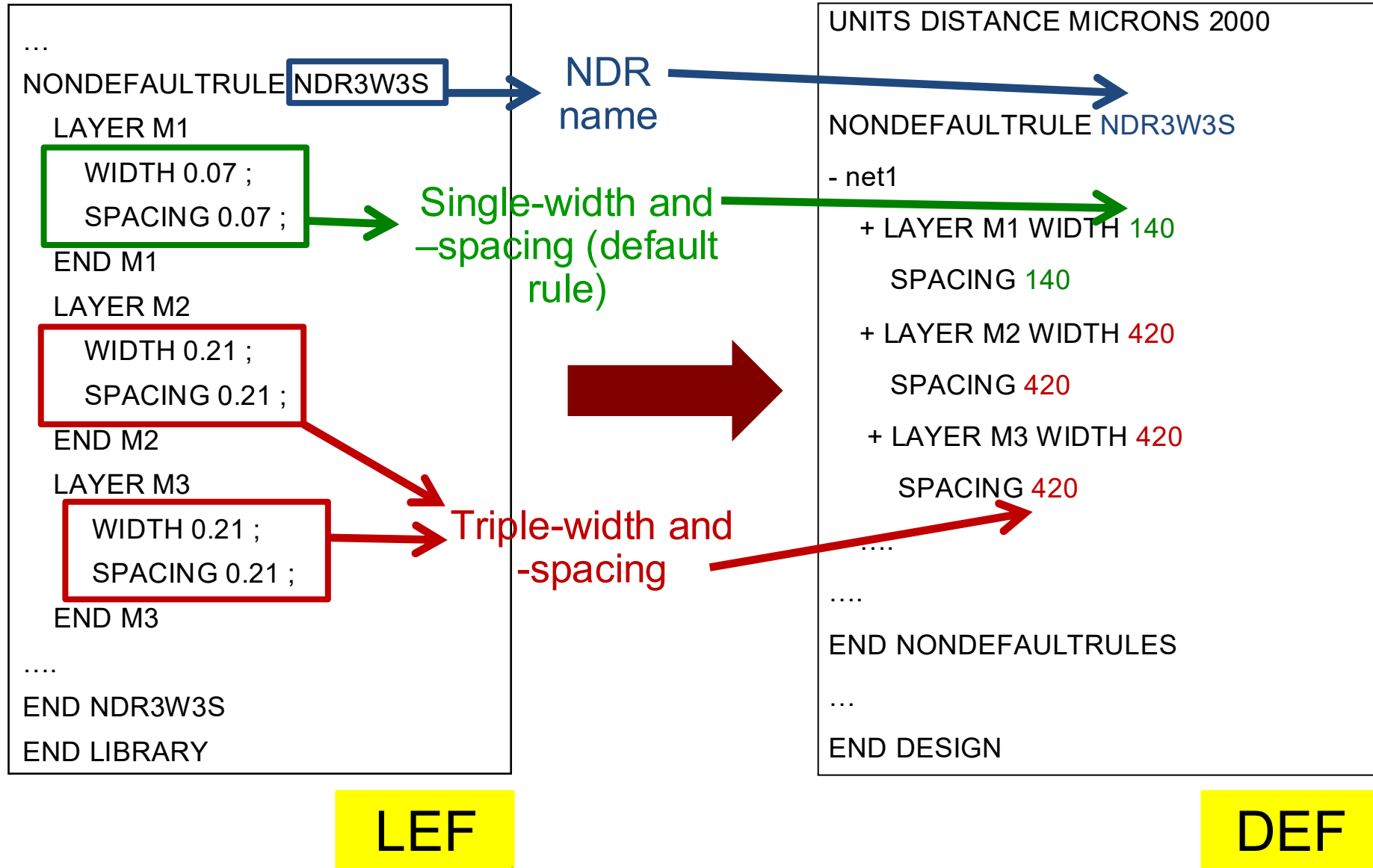
Multi-Modal Variability in Spacer-Assisted Double-Patterning

Different variabilities for different realizations of a wire ! → When life gives you lemons ... !

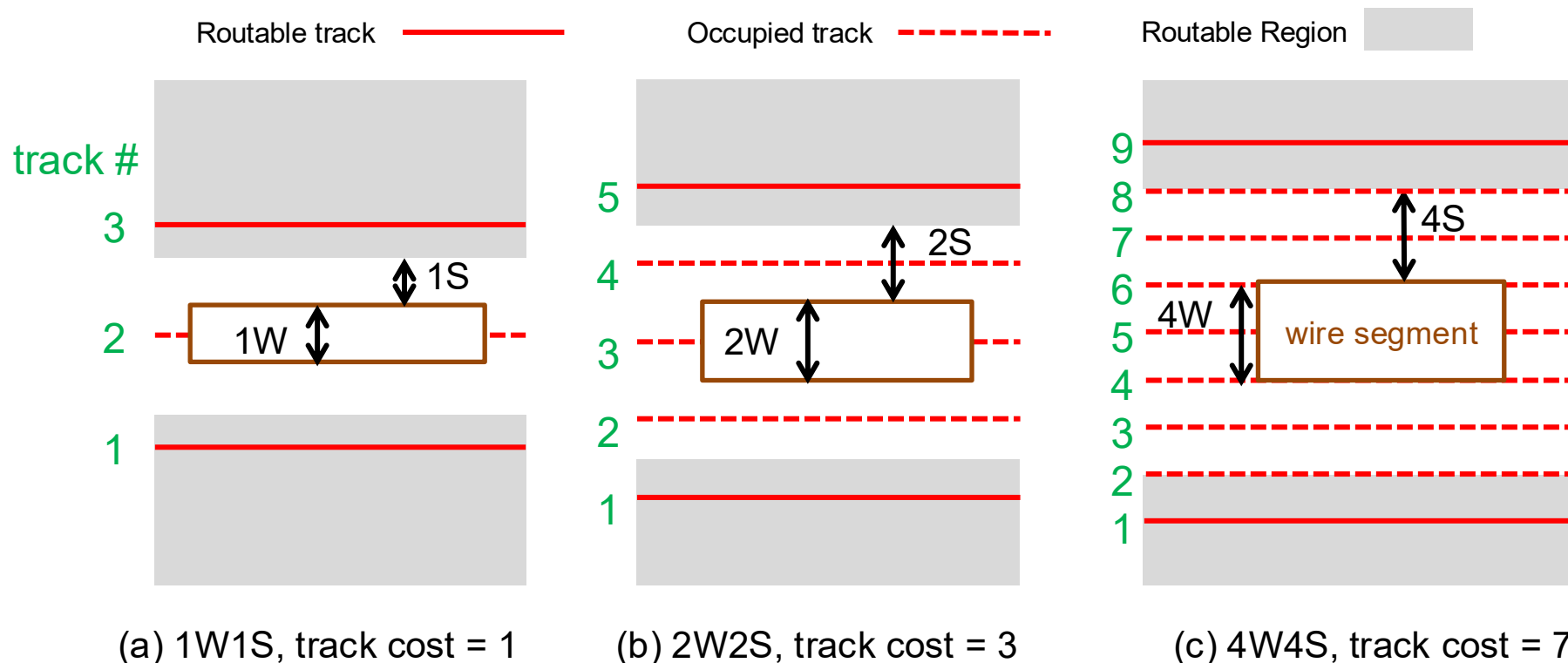


Feature		Variance (σ^2)	σ (nm)
Metal CD by mandrel		s_M^2	2.0
Metal CD by gap		$s_M^2 + (2s_S)^2$	3.3
Metal CD by wide mandrel+block		$(0.5s_M)^2 + s_{M-B}^2 + (0.5s_B)^2$	8.1
Metal CD by wide gap+block		$(0.5s_M)^2 + s_S^2 + s_{M-B}^2 + (0.5s_B)^2$	8.2

A Form of “Spreading”: Specification of NDRs in LEF, DEF



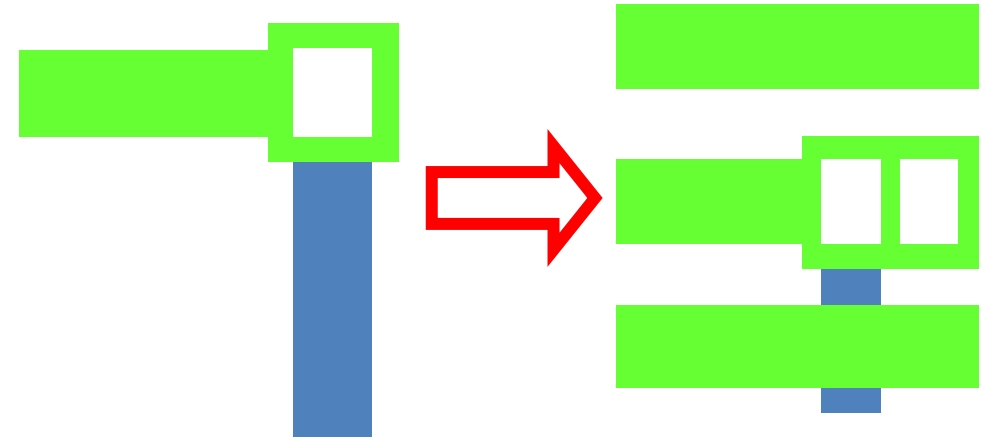
Track Impact of NDRs



- Track cost = Number of routing tracks unavailable to the detailed router because of increase in the required spacing to neighboring wires when applying NDRs
- Key: Router centers each wire segment on a track gridline**

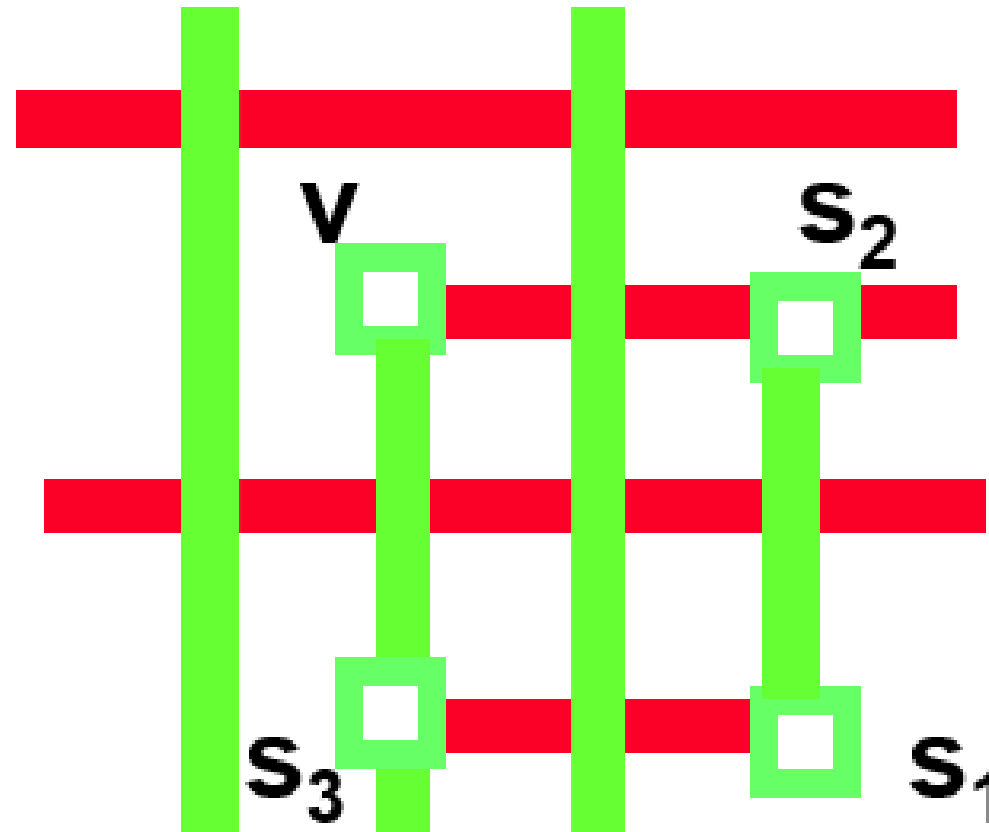
Via Doubling For Redundancy

- Single-cut via →
Double-cut via (“via doubling”)
 - During Routing?
 - or After Routing?
- Why redundant vias?
 - Improve yield: redundant via is “insurance” for via opens
 - Improve timing yield: the resistance of partially blocked vias increases
 - Reservoirs of metal for electromigration reliability
 - Etc.
- Observation:
 - The first 70-80% of vias can be easily doubled even for very congested designs



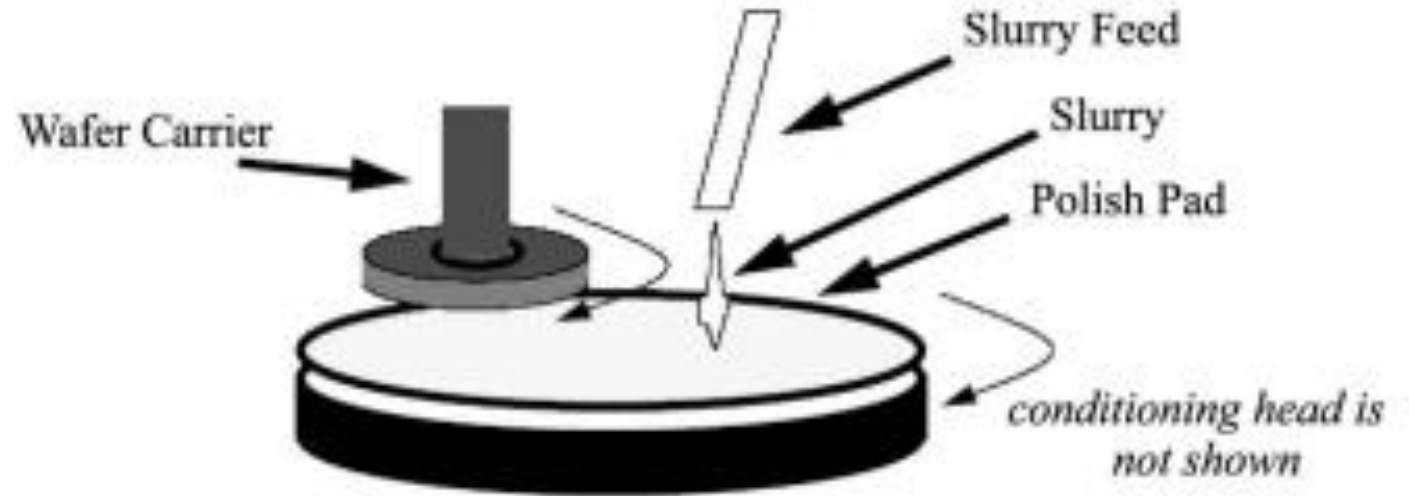
Short-Loop Paths to Add Redundancy

- Short-loop paths
 - Create detours to double vias which cannot otherwise be doubled with an adjacent single via (figure: via “v” is made redundant with a short loop)
 - Up to **97.5%** via doubling coverage by IBM Boeblingen in early 2000's
 - May lead to antenna / timing issues



Completion Criteria: *Demonstration of rules-compliant layout finishing functions including TCD, well taps, antenna checks, GDS merge, dummy fill.*

Dummy Fill for CMP Uniformity



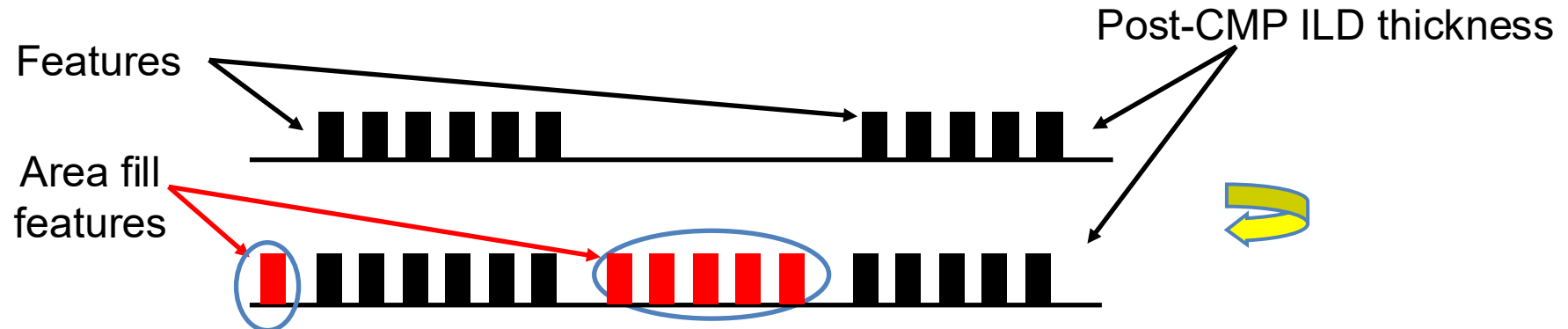
<https://vlsicad.ucsd.edu/Publications/Journals/j40.pdf>

<https://vlsicad.ucsd.edu/Publications/Journals/j84.pdf>

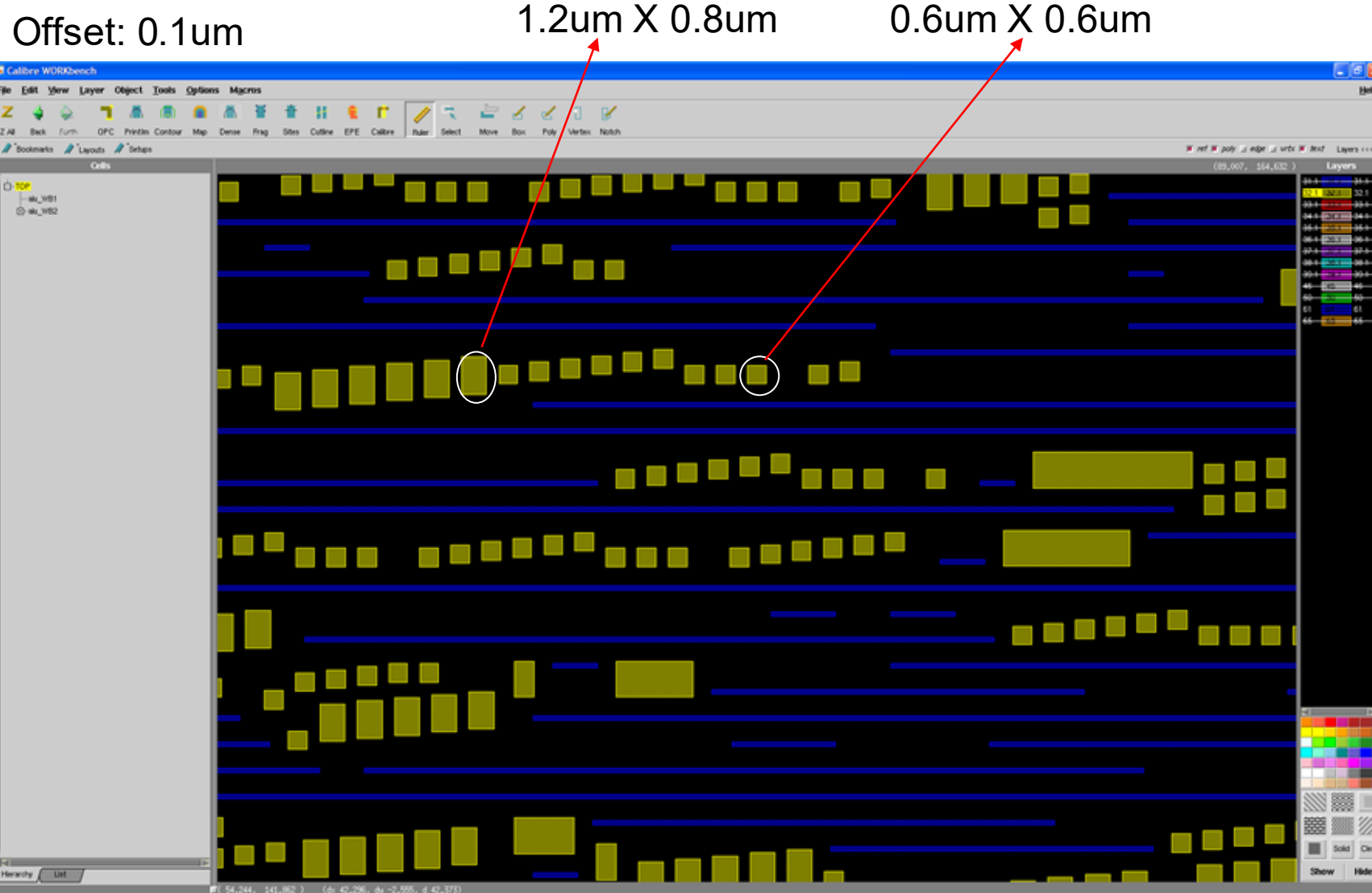
Manufacturability: Layout Density Control (“Fill”)

- Area fill or “dummy fill”: electrically inactive; floating or grounded
- Area fill insertion (**and, slotting**)
 - Decreases local density variation (□ area density design rules)
 - Decreases post-CMP interlayer dielectric (ILD) erosion, conductor dishing
 - CMP: Chemical-Mechanical Polishing (Planarization)
- Cf. “Filling and Slotting: Analysis and Algorithms”, [ISPD-98](#)

Must satisfy min, max area density limits in all windows of given size and “step” (e.g., density between 20-80% in 100um x 100um windows, with step size = 25um)

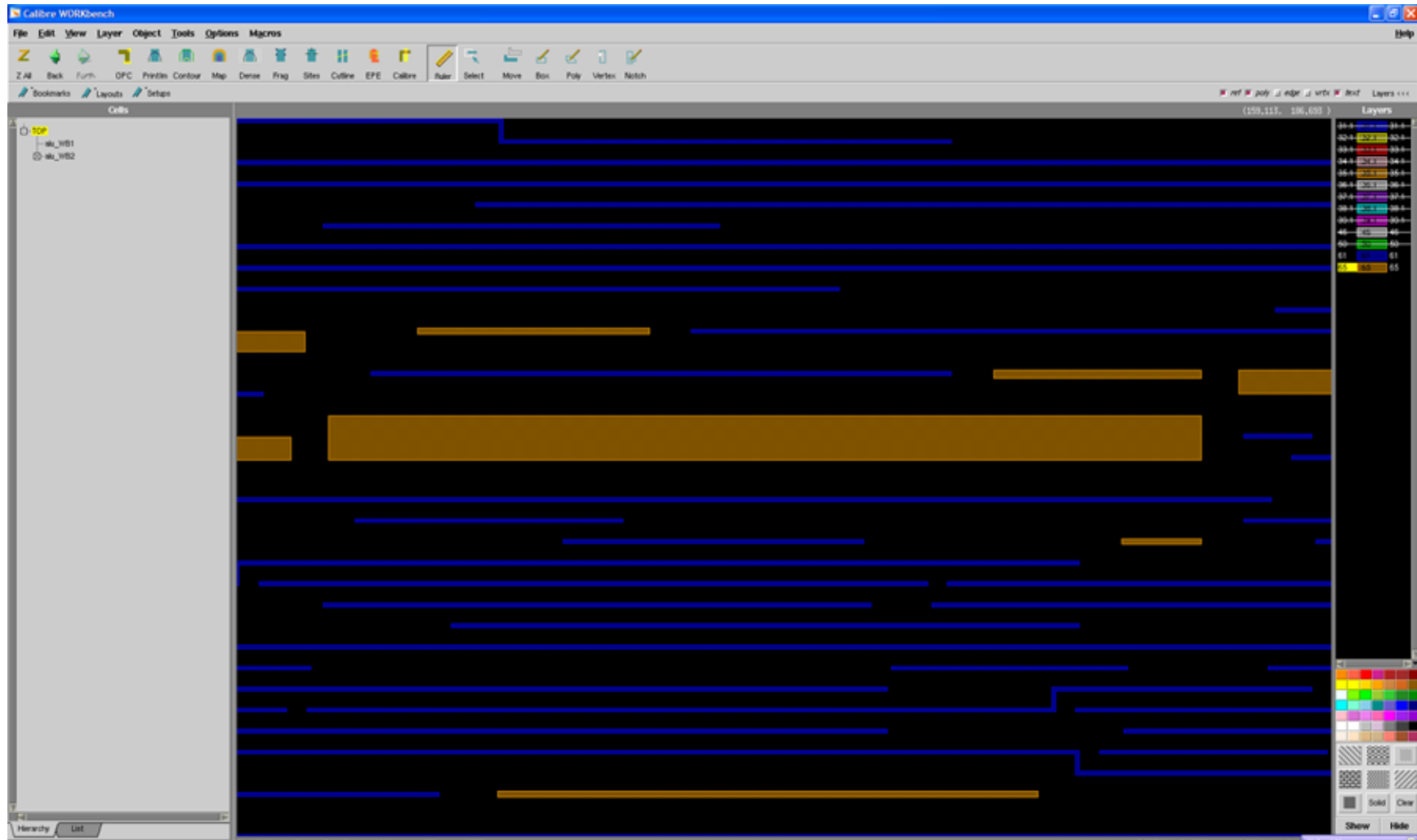


Calibre TSMC90 Fill (multipass, staggered)



“Tilted” or “staggered” dummy fill: Why?

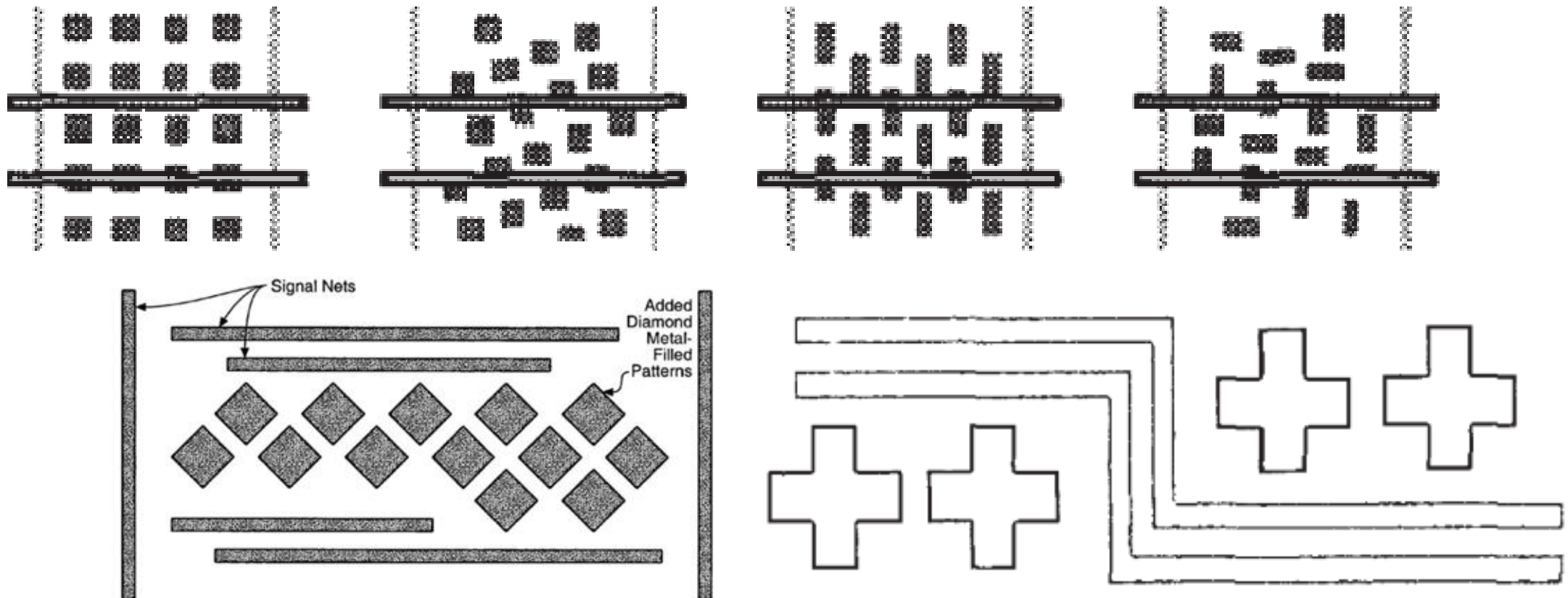
SOC Encounter TSMC90 Fill



“Track-like” dummy fill: Why?

Other Examples <https://vlsicad.ucsd.edu/Publications/Journals/j84.pdf>

- Top (l-r): traditional, staggered, alternating, basket-weave why?
- Bottom: diamond and plus-sign patterns



OpenROAD: “max”, “fixed pattern” (types)

<https://openroad.readthedocs.io/en/latest/main/src/fin/README.html#example-scripts>

- “fin” = layout finishing
- Metal fill can be part of output gds (foundry can insert fill as well)

The rules `json` file controls fill and you can see an example [here](#).

The schema for the `json` is:

```
{
  "layers": {
    "<group_name>": {
      "layers": "<list of integer gds layers>",
      "names": "<list of name strings>",
      "opc": {
        "datatype": "<list of integer gds datatypes>",
        "width": "<list of widths in microns>",
        "height": "<list of heights in microns>",
        "space_to_fill": "<real: spacing between fills in microns>",
        "space_to_non_fill": "<real: spacing to non-fill shapes in microns>",
        "space_line_end": "<real: spacing to end of line in microns>"
      },
      "non-opc": {
        "datatype": "<list of integer gds datatypes>",
        "width": "<list of widths in microns>",
        "height": "<list of heights in microns>",
        "space_to_fill": "<real: spacing between fills in microns>",
        "space_to_non_fill": "<real: spacing to non-fill shapes in microns>"
      }
    }, ...
  }
}
```

SKY130 M5 example: fixed pattern
= fixed density of 42.5%
(how did I see this?)

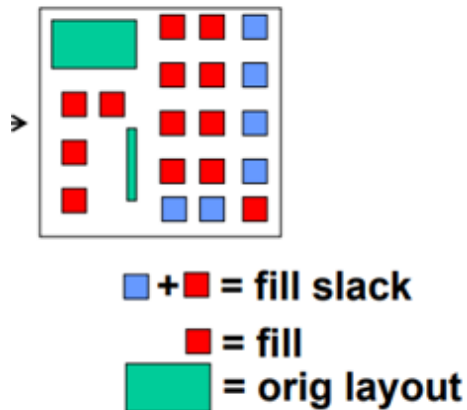
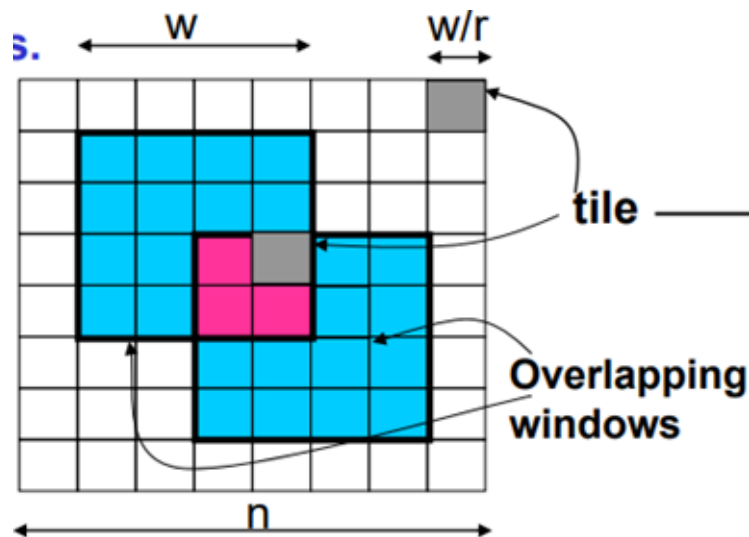
```
63 "met5" : {
64   "layer": 59,
65   "name": "met5",
66   "dir": "H",
67   "datatype": 0,
68   "space_to_outline": 70,
69   "non-opc": {
70     "datatype": 0,
71     "width": [3.00],
72     "height": [3.00],
73     "space_to_fill": 1.6,
74     "space_to_non_fill": 3.0
75   }
}
```

“Smart Fill”

<https://vlsicad.ucsd.edu/Publications/Journals/j40.pdf>

- Note: **many** ways that “dummy fill” can be smarter
 - Less data volume (GDS, OASIS compressibility)? Friendly to IP reuse? Smooth density gradient? Grounded fill for PI/SI? Timing-aware,-fixing? Control both area and perimeter of metal layout? (see papers at <https://vlsicad.ucsd.edu>)
- One key type of “smarter”: don’t add a maximum amount of fill !!!
 - Capacitance = delay and power; data volume considerations as well
- “Optimal” fill: minimize “**M**” in the linear program

<https://vlsicad.ucsd.edu/courses/cse101-w18/slides-w18/cse101-w18-Lecture13-linearprog.pdf>



$M \geq |\text{dens}(W_i) - \text{dens}(W_j)|$ for all windows i, j
// force minimum variation of post-fill window densities
 $\text{dens}(W) = \text{density}(\text{orig layout} + \text{fill})$ in all tiles of W
// density of a window = original + added fill
 $0 \leq \text{fill}_{ij} \leq \text{slack}_{ij}$ for all tiles T_{ij}
// cannot put more fill than will fit (i.e., more than “slack”)
// or negative amount of fill into any window

“Fill slack” = maximum amount of fill that can possibly be added into a given “file”

Conclusion

- “Layout Finishing” required by DARPA’s task definition in the IDEA program
- “Task 12” scope was addressed, but many extensions and improvements remain
- DFM (eDFM, DAM+MAD) = high-value **mindset** going forward (scaling slowdown, etc.)

How should design comprehend the process so that more value is extracted from it ?

OpenRCX

Thanks: Dimitris Fotakis and David Overhauser

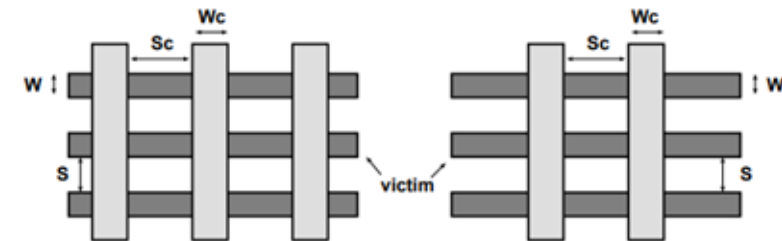
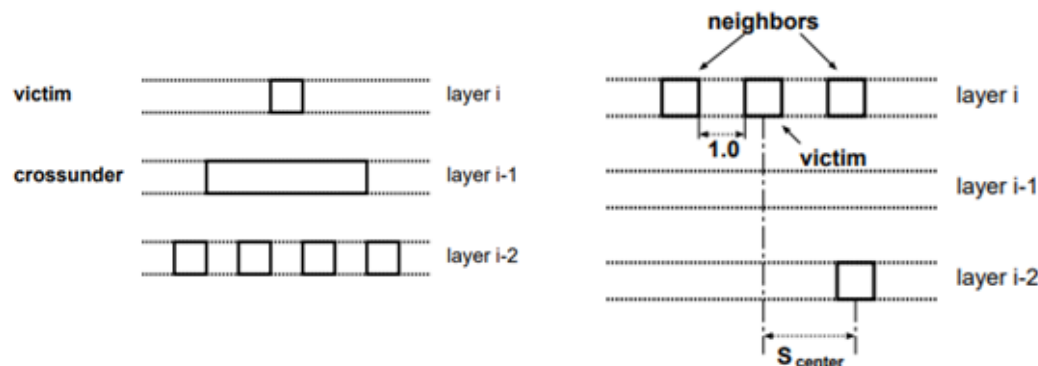
Capacitance Extraction

- What is distribution of charge on conductor surfaces and in dielectrics?
 - Solve Maxwell's equations $\rightarrow$ electric field and potential everywhere
- $C = Q/V \rightarrow$ obtain capacitance values
- Slow and accurate: 3D field solvers
- Quick and dirty: 1D (per-unit length total cap)
or 2D (per-unit area cap + per-unit length lateral, fringing cap)
- Good in practice: 2.5D $\approx$ 2D + corrections for adjacent layers
- OpenRCX is a 2.5D method
- Key challenge: building the (lookup table, interpolation) model for “corrections ...”

DAC 1997: “Five Foundations”

<https://vlsicad.ucsd.edu/Publications/Conferences/69/c69.pdf>

- Foundations (1997) for lookup tables and interpolations
 1. Shielding effects of ground, and same-layer neighbors
 2. High metal density on Layer i (e.g., with dummy fill !) decouples Layers $i-1$, $i+1$
 3. Layers $i \pm 2$ can be treated as ground planes; no need to look beyond
 4. Left/right same-layer couplings can be treated independently, and neighbor widths have little impact
 5. Above/below crossunders can be treated independently
- Systematic design of experiments (patterns), use of field solvers



Can Go Much Deeper (e.g., Fill Impact)

<https://vlsicad.ucsd.edu/Publications/Conferences/205/c205.pdf>

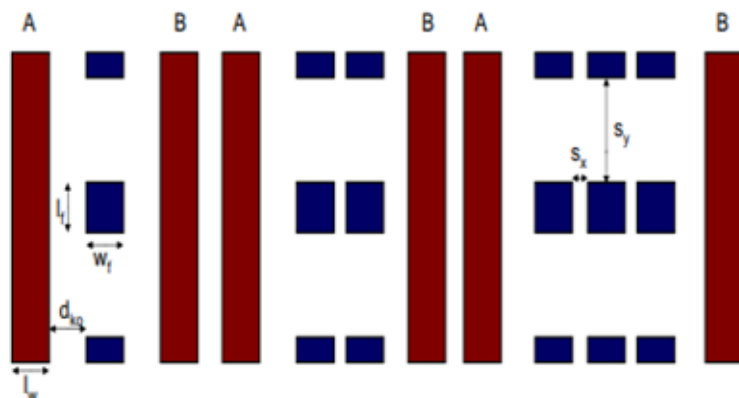


Fig. 12. Floating fill pattern examples.

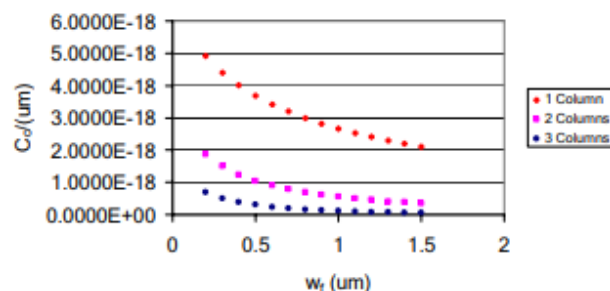


Fig. 15. Variation of coupling capacitance between active interconnects with respect to fill width.

IV. IMPACT OF CMP FILL ON INTERCONNECT PERFORMANCE

Chemical-mechanical planarization enables multilayer interconnect architectures. To enhance uniformity of post-CMP wafer topography, dummy fill is inserted to improve the uniformity of (effective) feature density. However, while dummy fill insertion improves feature density uniformity, it also changes coupling and total capacitance of functional interconnects [7], [11], [12].

In this section, we first study the impact of floating and grounded metal fill shapes on coupling (C_c) and total (C_{tot}) interconnect capacitance, in the context of various metal and fill configurations. We then motivate and outline an “intelligent fill synthesis” methodology that may be appropriate in future technology nodes.

A. Modeling of Fill Patterns

In the following discussion, the following notation is used.

- w_m : Line width
- h_m : Metal height
- h_{fd} : Dielectric height
- l_f : Fill length
- w_f : Fill width
- h_f : Fill height
- s_x : Horizontal spacing between fill features
- s_y : Vertical spacing between fill features
- d_{ko} : Keep-out distance

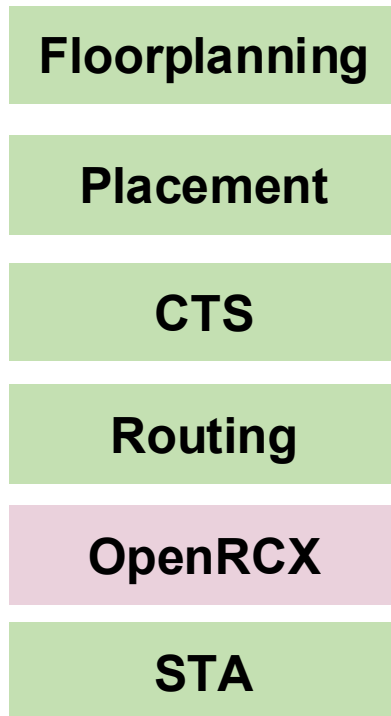
In Figure 20 we sketch a practical approach to *intelligent* timing-driven fill.

Timing-Driven Fill
Loop: 0. Set an initial conservatism factor 1. Do (initial) RCX and STA 2. Identify timing-violating nets (TVNs) - i.e., timing-critical nets 3. Apply conservative net-protection (+keep-out distance and blocking $M + 1/M - 1$ layers) per TVN segment 4. Run (incremental) MC-Fill $\rightarrow$ target fill amount 5. PIL-FILL Synthesis: 5.1 Greedy insert fill in fill slack columns, targeting most-needy tiles and largest and largest-slack nets first 5.2 After K fill shapes have been inserted, re-run (incremental) STA based on ΔC's 5.3 Iterate until all required fill has been inserted (or, until no timing constraint looks safe) 6. Update Conservatism 6.1 Analyze windows that violate <i>min</i> constraints 6.2 Identify nets that belong to the windows that violate the constraints 6.3 Do (incremental) RCX and STA to change the conservatism factor of TVNs (return to Step 2)

Fig. 20. Timing-driven fill synthesis approach.

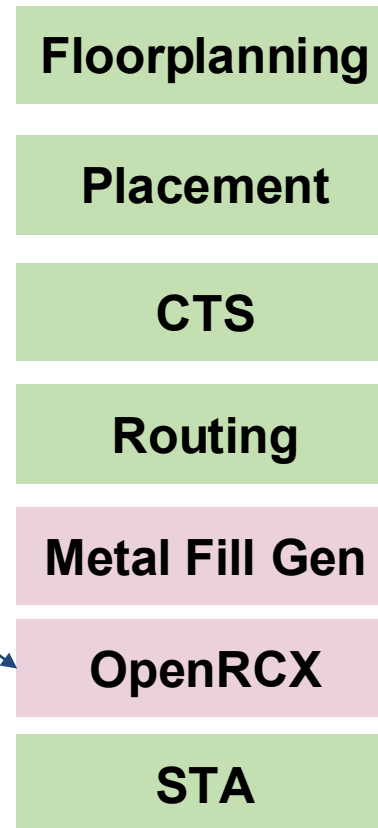
OpenRCX in the Design Flow

No Metal Fill Flow



Model File (1) - No Metal Fill including in Pattern Modeling

Metal Fill Flow



Note 1: if metal fill is present in the design it will be extracted as grounded wires.

Note 2: The challenge is that metal fill consideration is much larger area than pattern size; existence of fill must be either explicit or built in the extractor

Model File (1)

OpenRCX Internal Flow

Read Design (LEF, DEF, ODB)

Net Connectivity

Read Model File

RC Graph Generation

Build Geometry Structures

Extract Resistance per Wire

Extract Capacitance per Wire